

Pancreatic cancer

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1 **Pancreatic cancer**

2 *Nature Reviews Disease Primers*

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36 **Abstract**

37 Pancreatic cancer is a major cause of cancer-associated mortality with a dismal overall prognosis that
38 has remained virtually unchanged for many decades. At present, prevention or early diagnosis at a
39 curable stage is exceedingly difficult, therapeutic options are limited, and its complexity at genomic,
40 epigenetic, and metabolic levels as well as the multilayered interplay of neoplastic and host cells in
41 the tumor microenvironment continue to be challenging. In this review, we summarize the current
42 understanding of the salient pathophysiological, molecular, translational, and clinical aspects of this
43 disease. In addition, we present an outline of potential future directions for pancreatic cancer
44 research and patient management.

1. Introduction

The majority of malignant neoplasms of the pancreas are adenocarcinomas; rare pancreatic neoplasms include neuroendocrine tumors, which may secrete hormones such as insulin or glucagon, acinar carcinomas which may release digestive enzymes into the circulation, and even rarer neoplasms such as pancreatoblastomas and solid-pseudopapillary neoplasms. Pancreatic cancer, the most common subtype of which is pancreatic ductal adenocarcinoma, is a substantial health problem with an estimated 367,000 new cases diagnosed worldwide in 2015 and an associated 359,000 deaths in the same year ¹. Pancreatic cancer is currently the fourth highest cause of cancer death in developed countries, and if outcomes are not improved, is predicted to be the second leading cause of cancer related mortality within the next decade ². Risk factors, such as tobacco smoking, type 2 diabetes mellitus, and chronic pancreatitis account for approximately one quarter to one third of cases.

Pancreatic cancer is associated with an extremely poor prognosis ³ for a number of reasons. It is usually diagnosed at advanced stages, often due to nonspecific and in some cases even absent symptoms, lack of sensitive and specific tumor markers, and difficulties in imaging early stage tumors. Pancreatic cancer displays an aggressive growth behavior with perineural and vascular local growth and early distant metastases precluding curative surgical resection in most cases. Pancreatic cancer is characterized by a remarkable resistance/tolerance to most conventional treatment options (chemotherapy, radiotherapy, targeted therapy). Finally, pancreatic cancer harbors multiple genetic and epigenetic alterations and a complex and dense tumor microenvironment.

All of these factors result in an overall five-year survival rate of less 7%, with almost all of the survivors originating from the 10-20% of patients who undergo operative resection among whom the 5-year survival rate is approximately 20%. Although there has been incremental progress in some areas, such as more effective chemotherapeutic options, new preoperative treatment strategies, and better perioperative care/safer surgery, the overall effect on the prognosis of pancreatic cancer has been marginal. Novel strategies to treat pancreatic cancer include identifying subgroups for individualized and more effective therapies, targeted therapies, immunotherapies and focusing on the microenvironment as a potential target. Additional areas of intense investigation are early detection, tumor marker validation, and standardization of care.

In this primer, we will summarize key aspects of pancreatic cancer, from a basic, translational and clinical point of view, and provide an outlook of the most important challenges to tackle this deadly disease.

2. Epidemiology

When considering population based data, one must realize that there are geographical and temporal variations in the quality of clinical diagnoses and in the proportion of histologically verified cases of pancreatic cancer, which is still rarely over 50%⁴. Even when comparing populations living in the same place at the same time (e.g., different social classes or age groups), differential access to health care may influence the accuracy of reported pancreatic cancer rates⁵.

It is estimated that there have been 367,000 new cases of pancreatic cancer diagnosed worldwide in 2015, over 50% of which occurred in high-income countries¹. The highest pancreatic cancer rates were registered among Blacks in the USA (12-15/100,000 in men and 8-10/100,000 in women) and among indigenous populations in Oceania. In the USA, rates are about 30-50% higher in Blacks compared to Whites⁶. The lowest rates, likely influenced by under-diagnosis, were recorded in India, and Northern and Central Africa (below 2/100,000 in men and 1/100,000 in women). Mortality rates closely parallel incidence rates (figure 1). Increases in reported incidence and mortality have taken place since the 1970s, in particular in high income countries where it can be attributed in part to diagnostic improvements. However, incidence and mortality have levelled off over recent years in Western Europe and North America^{7,8}. Urban populations tend to have higher rates than rural ones, likely reflecting quality of diagnosis although there may also be differences in risk factors. Migrant studies suggest that first-generation migrants from low- to high-risk areas experience, after 15 or 20 years, rates similar to those of the country of migration⁹.

Risk factors

Age is the biggest risk factor for pancreatic cancer. Most cases occur over the age of 50, with peak incidences in the 7th and 8th decades of life. The best known preventable risk factor for pancreatic cancer is tobacco smoking. The risk in smokers is 2 to 3 fold higher than that in non-smokers, with a dose-risk relationship and a favorable effect of quitting smoking. The proportion of cases of pancreatic cancer attributable to tobacco smoking has been estimated to be 15% to 30% in various populations^{10,11,12}. Use of smokeless tobacco products may also increase the risk¹³. Being overweight/obese and low physical activity are linked to pancreatic cancer^{14,15}. Some nutritional and dietary factors have been related to pancreatic cancer, including high intake of (saturated) fats, low intake of vegetables and fruits, and consumption of red and processed meats¹⁶⁻¹⁸. Other dietary factors are however, largely undefined and unquantified. Early reports of an association between coffee consumption and pancreatic cancer risk were flawed and have not been confirmed by larger, more recent investigations. A positive association between heavy alcohol consumption and pancreatic cancer has been observed but there is no association with moderate drinking^{19,20}. This may be related to chronic pancreatitis, which increases the risk more than 10-fold, with little

114 difference in attributable risk between the alcoholic and non-alcoholic forms of the disease. Diabetes
115 mellitus appears to be both a risk factor for disease and a consequence of early pancreatic cancer.
116 Long-term diabetes mellitus approximately doubles the risk, and the risk is higher in the three years
117 following a diagnosis of diabetes mellitus ²¹. Diabetes mellitus can be caused by pancreatic cancer,
118 and new onset diabetes mellitus can be the first clue to the diagnosis in elderly patients ^{21, 22}.
119 Gastrointestinal ulcer and gastrectomy are associated with some modest increased risk of pancreatic
120 cancer, but this has however limited impact on overall disease burden ²³.
121 Ten percent of patients report a familial history of pancreatic cancer ²⁴. Specific hereditary
122 conditions carrying an increased risk of pancreatic cancer such as for example the Peutz-Jeghers
123 syndrome are summarized in table 1. Genes associated with pancreatic cancer include high-
124 penetrance genes such as *BRCA2*, *p16/CDKN2A*, *ATM*, *STK11* and *PALB2*, and common variants, which
125 confer to a modest risk, such as those at the ABO blood group locus ²⁵.

126

127 **3. Mechanisms & Pathophysiology**

128 Carcinogenesis, molecular biology, and signaling

129 Pancreatic cancer most often arises from pancreatic intraepithelial neoplasia (PanIN), the classical
130 pre-neoplastic lesions that may derive from regions of acinar-to-ductal metaplasia (ADM), and
131 pancreatic cancer may also arise from larger precursor lesions called intraductal papillary mucinous
132 neoplasms (IPMNs) and mucinous cystic neoplasms ²⁶. Invasive pancreatic cancer exhibits aberrant
133 autocrine and paracrine signaling cascades that promote pancreatic cancer cell proliferation,
134 migration, invasion, and metastasis. For example, many tyrosine kinase receptors such as the
135 epidermal growth factor receptor (EGFR), human EGFR 2 (HER2), HER3, insulin-like growth factor
136 receptor (IGF1R), fibroblast growth factor receptors (FGFRs) and the hepatocyte growth factor (HGF)
137 receptor (c-Met), and ligands such as transforming growth factor alpha (TGF- β), IGF-1, FGFs, and
138 HGF, activate multiple signaling cascades that enhance pancreatic cancer cell mitogenic self-
139 sufficiency and promote migration and invasion ²⁷. These occur in conjunction with the activation of
140 anti-apoptotic and pro-survival pathways such as STAT3, NFkB, and AKT ²⁷. Genes normally active
141 during development, such as Wnt, sonic hedgehog, and Notch, are reactivated in some pancreatic
142 cancers ²⁸.

143 Signaling in pancreatic cancer is complex, consisting of multiple nodes and aberrant cross-
144 talk pathways. For example, EGFR homodimerizes but also heterodimerizes with HER2 and HER3, and
145 in the context of oncogenic KRAS (see mutational landscape) and overexpression of multiple ligands
146 that bind EGFR and HER3, downstream signaling is excessively activated ²⁹. Similarly, c-Met signaling
147 is enhanced due to elevated expression of neuropilin 1 (NP1), integrins, and CD44 variants, all of
148 which interact with c-Met to further promote HGF's ability to stimulate migration and invasion ³⁰.

Moreover, elevated integrin- β 1 levels in pancreatic cancer can enhance EGFR signaling³¹. These observations point to an aberrant signaling nexus in pancreatic cancer that consists of Met/EGFR/NP1, CD44, and ITG- β 1, aggravated by c-Met's ability to heterodimerize with EGFR³². These alterations occur in the context of oncogenic *KRAS*, *p16/CDKN2A* tumor suppressor gene loss, and high levels of HGF and/or multiple ligands that bind to EGFR family^{26, 27}. Some of these ligands are heparin-binding, and their signaling becomes more sustained in the presence of the heparan sulfate proteoglycan glypican-1 which is overexpressed in pancreatic cancer^{33, 34}, highlighting a cascade of events that can promote pancreatic progression and metastasis (figure 2).

Pancreatic cancers also exhibit metabolic abnormalities and insensitivity to growth inhibitory pathways. Loss of negative growth constraints is best exemplified by aberrant transforming growth factor beta (TGF- β) signaling, which occurs in the face of increased expression of TGF- β isoforms³³. Although TGF- β is a physiological tumor suppressor, it promotes tumor progression in pancreatic cancer and many other solid tumors by exerting paracrine effects within the tumor microenvironment (see microenvironment) that lead to enhanced growth and metastasis. TGF- β can also directly induce pancreatic cancer cell proliferation by activating non-canonical signaling through phospho-ERK, src, and phospho-AKT, and by upregulating Wnt7b expression through canonical Smad4-dependent mechanisms³³.

Thus, in contrast to the situation in the normal pancreas where acinar and ductal cells are not exposed to excessive signaling, inflammation, gene mutations, or reactive stroma (figure 3), the cancer cells in pancreatic cancer are subjected to complex signaling events rendered even more aberrant by a plethora of driver mutations and cross-talk pathways, further impeding therapeutic interventions.

Mutational landscape and subtypes

The molecular pathology of pancreatic cancer is dominated by activating mutations of *KRAS*, present in over 90% of cases. Inactivating mutations of *TP53*, *CDKN2A* (p16) and *SMAD4* are present in 50 – 80% of pancreatic cancers, with a further handful at approximately 10% prevalence including *ARID1A*, *MLL3*, *TGFBR2* and *ATM* (figure 4). These are mere islands in an ocean of infrequently mutated genes that mostly occur at a prevalence of <2%³⁵⁻³⁹ (reviewed in⁴⁰).

Whilst point mutations of individual genes reveal some aspects of disease pathophysiology, other genomic events contribute to carcinogenesis and can provide insights into molecular mechanisms. For example, *CDKN2A* and *SMAD4* are commonly inactivated through homozygous deletion or methylation. Copy number alterations, which can be difficult to interpret due to the large number of genes that are amplified or lost in these regional events, also appear to also play an important role. Four types of pancreatic cancer have been hypothesized based on the number and

184 location of structural variants³⁸ (figure 4). The first three types include a stable genome with few
185 structural variants (<50), scattered structural events (50 – 200), and an unstable genome (>200
186 structural variants). The latter pattern is suggestive of defects in DNA maintenance and a potential
187 biomarker for platinum and poly (ADP-ribose) polymerase (PARP) inhibitor responsiveness³⁸. The
188 fourth group, termed locally rearranged is defined by a large number of events localized to 1 - 3
189 chromosomes. These are typically either amplifications which contain oncogenes, that can be the
190 targets of existing therapeutics, or genomic catastrophes such as chromothripsis.

191 Whole genome sequencing also reveals patterns of mutation that reflect mutational
192 mechanisms that were operative during each cancer's development, including toxins such as tobacco
193 and ultraviolet (UV) light exposure⁴¹. Mechanisms of DNA damage that contribute to pancreatic
194 carcinogenesis can be identified using these analyses, and include ageing (deamination), aberrant
195 APOBEC activity and defective DNA maintenance (one being the BRCA mutational signature, the
196 other defective mismatch repair). Interestingly, although smoking is associated with an increased risk
197 of pancreatic cancer, signatures of tobacco exposure as seen in lung cancer are not present⁴².

198 Analyses of epigenetic⁴³ and transcriptomic^{44,45} characteristics have also provided valuable
199 insights into pancreatic tumorigenesis. Analyses of mRNA expression have also suggested molecular
200 subtypes with potential relevance to therapeutic responsiveness, with subtypes of "epithelium" and
201 "stroma" recently defined. Whilst different analyses often arrive at different classifications, the
202 epithelial subtypes have been suggested to diverge based on their similarity to the normal or
203 embryonic pancreas. Those that bear the least resemblance to pancreas appear to have a poor
204 prognosis. More recently, two types of stroma (normal and activated) have also been described
205 based on mRNA expression profiles⁴⁵, which may explain disparate findings concerning the role of
206 the stroma⁴⁶ (see microenvironment), provide insights into potential immunoregulatory therapeutic
207 strategies, and predict outcome^{47,48}.

209 Microenvironment

210 Pancreatic cancers characteristically have an abundant collagenous stroma⁴⁹ (see pathology). This
211 stroma (desmoplastic reaction) is composed of extracellular matrix (ECM) proteins - collagens,
212 fibronectin and laminin, non-collagenous proteins such as glycoproteins, proteoglycans and
213 glycosaminoglycans, and other factors that possibly mediate the interaction of cancer cells with the
214 ECM including growth factors, osteopontin, periostin and serine protein acidic and rich in cysteine
215 (SPARC). Cellular elements of the stroma comprise pancreatic stellate cells (PSCs, which produce the
216 collagenous matrix), infiltrating immune cells, endothelial cells and neuronal cells. The immune cell
217 complement in pancreatic cancer includes T cells (a majority being CD4⁺ Tregs), myeloid derived
218 suppressor cells, macrophages and mast cells⁵⁰. Overall the immune cell infiltration suggests an

219 immunosuppressive phenotype, even at the earliest stages of pancreatic cancer i.e. PanINs and
220 IPMNs (reviewed in ^{51, 52}). A growing body of evidence suggests that Tregs in the stroma play a critical
221 role in warding off the host's immune system. This becomes clinically important as advancing
222 knowledge of factors that mediate suppression of active anti-tumor immunity such as the ligand for
223 PD-1 (PD-L1) expressed on cancer cells, forms the basis of novel immunotherapeutic approaches
224 currently under study in this disease ⁵³ (see management).

225 Whether the extent of stroma influences clinical outcome remains debatable. The largest
226 study to date with 233 patients reports an association between the numbers of activated PSCs and
227 poor clinical outcome ⁵⁴, but this was not corroborated by two subsequent, smaller studies ^{55, 56}. The
228 differences in the findings are likely to be a result of differences in the assessment/calculation of
229 stromal activity as well as differences in the patient cohorts selected for study.

230 Nevertheless, an active bidirectional interaction between stromal PSCs and cancer cells has
231 been well demonstrated by both *in vitro* and *in vivo* approaches ⁵⁷, with each cell type stimulating
232 proliferation and migration functions of the other (figure 5). Moreover, PSCs can inhibit cancer cell
233 apoptosis thereby increasing their survival, and also facilitate the formation of a "cancer stem cell
234 niche" that has been suggested to play a role in chemoresistance as well as recurrence of the disease
235 ⁵⁸. Furthermore, PSCs can travel from the primary tumor to distant metastatic sites, where they may
236 aid the seeding and growth of metastatic cancer cells ⁵⁹.

237 Interactions of PSCs with other stromal cells have also been reported in recent studies. Thus,
238 PSCs increase proliferation and tube formation (a measure of angiogenesis) by endothelial cells, an
239 effect mediated by vascular endothelial growth factor (VEGF) and/or the hepatocyte growth factor
240 (HGF, secreted by PSCs) / c-MET pathway ^{59, 60}. PSCs may contribute to the well-known phenomenon
241 of immune evasion in pancreatic cancer, by sequestering tumor suppressive CD8+ T-cells in the
242 stroma ⁶¹, inducing apoptosis of T-cells via secretion of the β -galactoside binding protein, galectin-1
243 ⁶², stimulating migration of myeloid derived suppressor cells into the stroma ⁶³, inducing
244 degranulation of mast cells leading to the release of tryptase and IL-13, which cause PSC and cancer
245 cell proliferation ⁶⁴; and inducing cytokine production by macrophages leading to further activation
246 of PSCs ⁶⁵. In the presence of PSCs neurite growth towards cancer cells and invasion of neurons by
247 cancer cells is increased, an effect possibly mediated by the sonic hedgehog pathway ⁶⁶. Interestingly,
248 it has been suggested that PSCs may have a role in the new-onset diabetes mellitus (type 3c) of
249 pancreatic cancer via their inhibition of beta cell function in pancreatic islets ⁶⁷ (figure 5).

250 Although the weight of evidence to date indicates a facilitatory role for stromal PSCs in tumor
251 growth and metastasis, two recent studies (using genetic techniques ⁵⁶ or signaling pathway
252 inhibition ⁶⁸ to deplete myofibroblast numbers/function), have controversially suggested that the
253 stroma may have a protective role in pancreatic cancer. These discrepant findings serve as a

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255 reminder that the influence of the stroma may be highly dependent on context and timing e.g. during
256 early carcinogenesis, PSC-produced stroma may be the host's attempt to cordon off tumor cells, but
257 in later stages the cancer cells may be able to subvert stromal PSCs into cancer-permissive cells.

258

259 Metabolism reprogramming

260 Changes in cell metabolism are a hallmark of carcinogenesis. Adaptation to the microenvironment
261 and oncogenes are key drivers in this process. As outlined above, pancreatic cancer is characterized
262 by a severely hypoxic and nutrient-deprived microenvironment^{69,70} and by oncogenic *KRAS*
263 mutations in the vast majority of cases. Pancreatic cancer cells have adapted to survive in these
264 harsh conditions through various mechanisms, mainly driven by the hypoxia-inducible transcription
265 factor-1A (HIF-1A)^{71,72} and oncogenic *KRAS*. These adaptive changes also constitute potential targets
266 for therapeutic intervention.

267 In response to the hypoxic and nutrient-deprived microenvironment and oncogenic *KRAS*,
268 pancreatic cancer cells switch to aerobic glycolysis with increased lactate formation, which might in
269 turn be an important nutrient for components of the microenvironment and less hypoxic cancer cells
270⁷³ (figure 6). Glucose uptake and utilization is increased, partially as a result of oncogenic *KRAS*
271 induced GLUT1, HK-1/-2 expression and other mechanisms^{74,75}. Utilizing glucose for aerobic
272 glycolysis reduces oxidation of pyruvate in the tricarboxylic acid cycle and subsequent oxidative
273 phosphorylation, thereby limiting reactive oxygen species (ROS) production. Further, oncogenic *KRAS*
274 induces Nrf2 that activates a number of anti-oxidant pathways to maintain low ROS levels. Glutamine
275 uptake is also increased in pancreatic cancer as *KRAS*-driven cells depend on glutamine metabolism
276⁷⁶. In pancreatic cancer, glutamine is necessary for synthesis of proteins and nucleic acid and for
277 maintaining the redox state⁷⁴. Further, the hexosamine biosynthetic pathway, in which glutamine
278 and glucose are channeled, is activated, thereby increasing O-GlcNacylation of a number of proteins
279 that are important for survival under hypoxic conditions⁷⁷. Increased fatty acid synthesis from
280 glutamine-derived α -ketoglutarate instead of glucose-derived pyruvate is supported by hypoxia
281 induced enzymes involved in fatty acid synthesis^{78,79}. Pancreatic cancer cells frequently overexpress
282 these enzymes⁸⁰. To maintain their biomass needs, pancreatic cancer cells depend on autophagy⁸¹
283 and *KRAS*-dependent macropinocytosis⁸² (Figure 6).

284 Targeting specific metabolic adaptations is an emergent strategy for pancreatic cancer
285 (reviewed in⁷⁴). These strategies include blocking key regulators of the tumor metabolism such as
286 pyruvate kinase (PKM2), lactate transport, autophagy and others.

287

288

289 Disease models

290 Model systems of pancreatic cancer are widely available and have substantially improved our
291 knowledge of this disease. Contemporary pancreatic cancer models complement the traditional cell
292 line and xenograft models⁸³, and include genetically engineered mutant mice (GEMM)⁸⁴⁻⁸⁶, and
293 organoid cultures⁸⁷⁻⁸⁹.

294 Mouse models of pancreatic cancer somatically target mutant alleles to the murine pancreas.
295 Such autochthonous models accurately mimic the human disease and have confirmed causative roles
296 for many genes previously implicated in the human pancreatic cancer genome. Indeed, GEMMs
297 revealed that oncogenic *KRAS* is uniquely sufficient to initiate Pancreatic Intraepithelial Neoplasms
298 (PanIN) – the presumed main preinvasive precursor of pancreatic cancer^{84,85}, and that such mice
299 spontaneously progress to locally invasive and metastatic pancreatic cancer. When combined with
300 oncogenic *KRAS*, additional orthologous mutations in the canonical tumor suppressor genes *P16*,
301 *TP53*, or *SMAD4* confirmed that they accelerate pancreatic tumor progression of varying histology⁸⁶,
302⁹⁰⁻⁹⁴, and this is the standard method to query the function of a potentially pathogenic allele.
303 Furthermore, novel pancreatic cancer genes and pathways can also be discovered using insertional
304 mutagenesis strategies in sensitized mice⁹⁵⁻⁹⁷.

305 Genetic requirements for pancreatic cancer progression can be established by incorporating
306 concomitant germline or conditional alleles into traditional models, and this has revealed the
307 importance of various signaling pathways including *Egfr*⁹⁸, *Pdk1*⁹⁹, and *Nrf2*¹⁰⁰ in tumor initiation.
308 Likewise, to assess the role of genes in tumor maintenance, flexible GEMMs using ligand-regulated
309 reversible alleles have been employed to confirm the dependency of pancreatic cancer on
310 continuous expression of oncogenic *KRAS*^{75,101} and loss of *TP53*¹⁰². For example, the silencing of
311 oncogenic *KRAS* has identified new metabolic alterations in pancreatic cancer that offer therapeutic
312 opportunities^{75,76}.

313 Cell biological processes involving neoplastic and microenvironmental interactions can also
314 be carefully probed using GEMMs, and such studies have surprisingly revealed that in mice acinar
315 and endocrine cells are more capable of initiating PanIN than ductal cells^{85,103-106}, and have
316 suggested that in mice histologically early stage PanIN cells are capable of local invasion and
317 intravascular dissemination⁶⁸. Additionally, GEMMs have identified a potent stimulatory role for
318 pancreatic inflammation⁸⁵ and myeloid cells^{107,108} in pancreatic cancer pathogenesis. GEMMs have
319 also demonstrated that pancreatic cancer differentiation is influenced by the juxtaposition of
320 activated pancreatic stellate cells^{56,68}. Finally, pancreatic cancer pathophysiological sequelae such as
321 cachexia¹⁰⁹ and pain¹¹⁰, and the role of obesity¹¹¹ and the intestinal microbial flora¹¹² can also be
322 evaluated in GEMMs.

323 Therapeutic studies with GEMMs directly prompted the realization that human pancreatic
324 cancer contained a markedly deficient vasculature ¹¹³, stimulating methods to increase intratumoral
325 perfusion and drug delivery. Although the clinical development of Hedgehog pathway inhibitors has
326 been disappointing ^{114, 115}, a number of analogous approaches using pegylated hyaluronidase ^{116, 117}
327 and synthetic Vitamin D analogues ¹¹⁸ are under active investigation (see management). Several
328 factors that suppress the immune response in pancreatic cancer have also been identified,
329 stimulating multiple clinical efforts ¹¹⁹⁻¹²². Finally, therapies that target the Ras pathway effectors ¹²³
330 and parallel pathways such as Notch ¹²⁴ have demonstrated some activity in GEMMs, again
331 motivating clinical development.

332 Mice harboring preinvasive neoplasms and advanced pancreatic cancer in various GEMMs
333 represent ideal systems to develop diagnostic methods. Accordingly, early detection strategies for
334 pancreatic cancer have been pioneered in GEMMs, including the discovery of the potential imaging
335 biomarkers Plectin 1 ¹²⁵ and Claudin 4 ¹²⁶, and blood biomarker panels ¹²⁷ and circulating exosomes
336 containing Glypican-1 ¹²⁸. Many of these have been corroborated in human samples, and clinical
337 confirmation is ongoing (see biomarkers).

338 To decrease mouse breeding times and costs, several strategies have been developed
339 including the mutagenesis of pancreatic cancer-sensitized ES cells followed by blastocyst injection
340 and chimera analysis ¹⁰², and the direct intra-pancreatic delivery of recombinant viruses harboring
341 gene editing constructs to rapidly generate pancreatic cancer models ¹²⁹. The near future should also
342 herald the development of multi-recombinase model systems to control temporal expression of
343 several alleles at different stages of tumor progression. These approaches require sophisticated
344 transgenic and surgical capabilities, and may be ideal for certain applications.

345 Recently, three dimensional organoid cultures representing human and mouse pancreatic
346 cancer have been described ⁸⁸. Organoids can be cultured indefinitely, molecularly characterized, and
347 subjected to drug screens. Furthermore, the orthotopic transplantation of human organoids
348 recapitulates some of the stages of tumor progression, thereby providing a model of early human
349 pancreatic cancer. As they can be readily established from small biopsies, organoids may reasonably
350 become a component of personalized medical approaches for pancreatic cancer patients. A current
351 goal is the establishment of a sufficient collection of human organoids to determine whether they
352 reflect the diversity of pancreatic cancer that was previously reported by sequencing tumors ^{35, 37, 44}.
353 In this regard, organoids may represent the heterogeneity of human pancreatic cancer better than
354 GEMMs, another area of active investigation (figure 7).

355 The GEMM and organoid models of pancreatic cancer can be used to represent the entire
356 spectrum of disease providing tractable preclinical and co-clinical experimental approaches for this
357 lethal malignancy. While each model has its limitations, these models will continue to be used to

358 identify new biological features of this disease - representing avenues for therapeutic and diagnostic
359 development.

360

361 **4. Diagnosis, screening and prevention**

362 Biomarkers

363 It may take more than a decade for a neoplastic cell in the pancreas to generate distant metastases
364 ¹³⁰providing a large window of opportunity for early detection ¹³¹. There is also the need for
365 biomarkers for response evaluation in the palliative and neoadjuvant setting as well as for post-
366 resection follow-up.

367 Several different types of diagnostic biomarkers may be useful in pancreatic cancer (Figure
368 8). The most widely used biomarker is CA19-9. It is a sialylated lacto-N-fucopentaose II related to the
369 Lewis (a) antigen. Unfortunately, CA19-9 lacks sufficient sensitivity or specificity to be useful for early
370 pancreatic cancer diagnosis. Similarly, the presence of circulating tumor cells (CTCs) derived from
371 pancreatic cancer could be diagnostic, but these cells are likely only present in patients with
372 metastatic disease. By contrast, circulating tumor DNA (ctDNA) was detected at the time of diagnosis
373 in 43% of patients with localized disease ¹³², suggesting that ctDNA could serve as an early diagnostic
374 test. Similarly, systemic metabolic alterations may be a harbinger of pancreatic cancer, as
375 demonstrated by the presence of elevated circulating branched amino acids (BCAA) in early stage
376 disease in a mouse model of pancreatic cancer and in some patients 2 to 5 years prior to their
377 diagnosis of pancreatic cancer ¹³³.

378 Recent studies point to novel biomarkers for early pancreatic cancer diagnosis. First,
379 antibody microarrays using sequential plasma samples from a genetically engineered mouse models
380 of pancreatic cancer and from pre-diagnostic plasma derived from the Women's Health Initiative
381 generated a highly specific protein signature consisting of estrogen receptor 1, ERBB2, and tenascin C
382 ¹³⁴. Second, the presence of the heparin sulfate proteoglycan glypican-1 on the outer layer of
383 circulating exosomes is seen with early stage disease, suggesting that this type of exosome analysis
384 could be useful for early diagnosis ¹²⁸. Exosomes also carry an internal cargo of proteins, nucleic
385 acids, and microRNAs and could yield additional novel diagnostic tests. For example, the presence of
386 adrenomedullin (AM) in exosomes from pancreatic cancer patients has been associated with
387 pancreatogenic (type 3c) diabetes mellitus, which may precede diagnosis by several months to a
388 couple of years ¹³⁵. Thus, assaying exosome-bound AM could help diagnose pancreatic cancer earlier.
389 Plasma microRNAs signatures ¹³⁶ may also be diagnostic for pancreatic cancer, and such assays could
390 be refined to allow for early pancreatic cancer diagnosis. Finally, circulating mesothelin has also been
391 proposed as a protein biomarker, and mesothelin-specific T cells can potentially be used to devise

immune interventions targeting this biomarker underscoring the fact that certain biomarkers can help establish novel therapies ^{137, 138}.

Diagnosis

The clinical manifestations of pancreatic cancer are non-specific, and include jaundice, unexplained weight loss, epigastric pain that radiates to the back, nausea, diabetes mellitus and, rarely, migratory thrombophlebitis ¹³⁹. When a diagnosis of a pancreatic cancer is suspected, the next step is imaging. Multidetector computed tomography (MDCT) provides excellent resolution of the pancreas and surrounding vasculature, and can be used to evaluate other organs for spread of the disease ¹³⁹ (figure 9). Most pancreatic cancers form solid hypodense lesions. Cancers of the head of the pancreas typically obstruct both the pancreatic and bile ducts, producing upstream dilatation of both ducts (the “double duct sign”) (figure 10), while carcinomas of the corpus and tail of the gland will only obstruct distal parts of the pancreatic duct. MDCT can not only be used to define lesions in the pancreas, but it also can be used to evaluate the resectability of any lesions. Involvement of the large vessels adjacent to the pancreas (such as the superior mesenteric artery and vein, figure 9), may make the tumor borderline resectable or locally advanced/unresectable (see surgery) ¹³⁹. Involvement of distant organs, such as the liver, would indicate advanced disease and that the patient is suitable for palliative therapy only; i.e. chemotherapy in most cases. Magnetic resonance imaging (MRI), can provide excellent resolution of the pancreatic ducts (figure 10) and of any cysts that might be present. MRI may also be more sensitive than MDCT in detecting and evaluating liver metastases. Since pathology is the gold standard in establishing the diagnosis of pancreatic cancer, most patients will undergo endoscopic ultrasound with fine needle aspiration biopsy (EUS/FNA) ^{140, 141}. EUS, although very operator dependent, provides excellent resolution of the pancreas and peripancreatic vessels and lymph nodes, and can be used to obtain tissue for definitive diagnosis. Positron emission tomography (PET) can supplement these technologies, particularly when evaluating enlarged lymph nodes and larger masses of uncertain clinical significance, such as a persistent mass lesion present after therapy ¹⁴². Serum cancer antigen 19-9 (CA19-9) levels (see biomarkers), although not useful for screening asymptomatic individuals, can be useful in managing patients with an established pancreatic cancer. CA19-9 levels in the thousands of IU/ml suggests metastatic disease, while a significant decline in levels after treatment suggests a good response. It should be noted, however, that 10-15% of the population lack the enzyme necessary to synthesize CA19-9, so measurement of CA19-9 levels will obviously not be useful in these individuals ¹⁴³. Other blood tests that are particularly useful in managing patients with pancreatic cancer, include serum bilirubin and fasting blood glucose levels. Biliary obstruction can cause hyperbilirubinemia, and paracrine effects by cancer and or stellate cells, as well as destruction of pancreatic parenchyma

427 including islets of Langerhans, can cause diabetes mellitus ¹⁴⁴. Indeed, new onset diabetes mellitus is
428 the first sign of pancreatic cancer in some patients ¹⁴⁵. Fasting blood glucose and hemoglobin A1C
429 levels can therefore be helpful in diagnosis. Improvements in technologies that can be used to
430 detect rare mutant fragments of DNA, coupled with the growing understanding of the genetic
431 alterations driving pancreatic neoplasia (see mutational landscape), suggest that DNA-based blood
432 tests for circulating tumor DNA may soon be useful tests for pancreatic cancer (see biomarkers) ¹⁴⁶.

434 Pathology

435 A number of non-neoplastic and neoplastic conditions can mimic pancreatic cancer clinically.
436 Pathology is therefore the gold standard in establishing a diagnosis. Depending on how the pancreas
437 is sampled, the materials obtained can be evaluated cytologically as smears, or histologically as tissue
438 sections ¹⁴⁷. As is true for other organs, the diagnosis of cancer is usually established by evaluating
439 nuclear features such as the shape, size and intensity of staining of nuclei. In addition to establishing
440 the diagnosis of cancer, pathology can be used to define the tumor's grade, an important
441 prognosticator, and tumor type.

442 Most pancreatic cancers are infiltrating "tubular" ductal adenocarcinomas. These are
443 characterized by gland-forming neoplastic cells infiltrating into an intensely desmoplastic stroma ¹⁴⁷
444 (figure 11). This desmoplastic stroma is important to recognize for two reasons. First, the stroma
445 can be so florid that small biopsies can miss the neoplastic glands. Extensive sampling is therefore
446 needed before a diagnosis of pancreatic cancer can be ruled out. Second, the desmoplastic stroma is
447 typically hypovascular and under significant hydrostatic pressure, potentially impeding the delivery of
448 chemotherapeutic agents to the neoplastic cells (see also microenvironment) ^{116, 148}.

449 Several distinct types of neoplasms can arise in the pancreas, and some of these have clinical
450 significance ¹⁴⁷. Pancreatic neuroendocrine tumors (PanNETs), are characterized, as the name
451 suggests, by extensive neuroendocrine differentiation. These neoplasms have characteristic nested
452 growth pattern, "salt and pepper nuclei," and express the neuroendocrine markers synaptophysin
453 and chromogranin. PanNETs tend to be slow growing, and are often best treated surgically. By
454 contrast, adenosquamous carcinomas and undifferentiated carcinomas, are usually fast growing and
455 are often widely metastatic. Some adenosquamous carcinomas, however, may respond to platinum
456 based chemotherapy ¹⁴⁹. Colloid carcinomas are characterized by the formation of pools of acellular
457 mucin in the stroma. Colloid carcinomas almost always arise in association with a distinct cystic
458 precursor lesion called an intraductal papillary mucinous neoplasm ¹⁴⁷. The prognosis for patients
459 with a colloid carcinoma is better than it is for patients with a typical ductal adenocarcinoma, but
460 some of the improved outcome may relate to the fact that these neoplasms tend to present at an
461 early stage. Finally, medullary carcinomas are a distinct form of cancer characterized by poor

462 differentiation, a syncytial growth pattern and pushing borders. Medullary carcinomas are important
463 to recognize because, despite their poor differentiation, they carry a good prognosis and often
464 display microsatellite instability (MSI). Cancers with MSI appear to be exquisitely sensitive to some
465 immunotherapies ¹⁵⁰. The precursor lesions that give rise to invasive pancreatic cancer include
466 pancreatic intraepithelial neoplasia (PanIN – graded from low to high grade dysplasia ¹⁵¹), intraductal
467 papillary mucinous neoplasms, and mucinous cystic neoplasms ¹⁴⁷.

469 Screening

470 Earlier detection of pancreatic cancer should theoretically lead to improved survival ¹³¹, although
471 whether this can be achieved in the context of a screening program is not yet well established. The
472 time from the initiating mutation to the birth of the parental founder cell is estimated to be at least a
473 decade ¹³⁰ and a number of curable non-invasive lesions that precede the development of pancreatic
474 cancer have been identified (see pathology). Detection and surgical resection of these lesions prior to
475 the development of an invasive cancer gives a high probability of cure ^{152, 153}. These lesions are,
476 however common, and there is a real risk of overtreatment. Improvements can also be made by
477 identifying pancreatic cancer at an operable stage, as surgical resection is associated with an
478 improved five-year survival from less than 5% to approximately 20% ¹⁵⁴. It has been estimated that
479 increasing the proportion of people diagnosed with stage I or II disease from 34% to 61%, with a
480 reduction in the proportion of patients diagnosed with stage III or IV disease, would result in a
481 doubling of survival ¹⁵⁵.

482 Population-based screening is not feasible due to the low lifetime risk (about 1.3% to age 70).
483 Population subgroups with at least a five-fold increased risk of developing pancreatic cancer may
484 benefit from screening. Efforts to stratify the general population on the basis of non-genetic risk
485 factors and single nucleotide polymorphisms have thus far failed to identify a group at sufficiently
486 high risk to warrant screening ¹⁵⁶. However, incorporation of prediagnostic symptoms into a risk
487 prediction model may identify a small subgroup of the population among whom screening could be
488 considered ¹⁵⁷.

489 Current screening investigations are limited to (1) people from familial pancreatic cancer
490 kindreds, who have up to a 32-fold increased risk depending on the number of relatives affected ¹⁵⁸,
491 or (2) people with mutations in genes known to increase risk, including *BRCA2*, *PALB2*, *CDKN2A (p16)*,
492 *STK11*, *ATM*, *PRSS1* genes and the hereditary colon cancer (Lynch syndrome) genes (table 1). The
493 yield of potentially actionable lesions identified in high-risk cohorts varies considerably according to
494 the inclusion criteria for the study, the screening methods used and the definition of an abnormal
495 lesion; in the largest series to date, with mean follow-up of 4.2 years, the yield of pancreatic cancer,
496 IPMN or PanIN of at least grade 2 was 7.3% ¹⁵⁹.

497 While there is general consensus that high risk individuals should be enrolled in a screening
498 trial, there is a paucity of evidence on which to decide the age at which screening should commence
499 and at which patients should be discharged from the screening program, the screening interval and
500 the optimal imaging modality. Current suggestions are that patients should begin screening at age 50
501 or 10 years younger than the age at which the youngest family member was diagnosed. At a summit
502 held by the International Cancer of the Pancreas Screening (CAPS) Consortium, approximately three-
503 quarters of experts agreed that magnetic resonance imaging or endoscopic ultrasound should be
504 used preferentially over computed tomography scan due to the higher yield, but there was no
505 consensus on screening interval ¹⁶⁰.

506 Screening carries risks of over-diagnosis and, apart from unambiguous solid lesions, it is
507 currently unclear which lesions should be managed surgically versus observed. For screened patients
508 who are found to have an IPMN It is probably appropriate to follow the current consensus guidelines
509 for the general population¹⁶¹. PanIN is difficult to detect and imaging cannot distinguish low- from
510 high-grade PanIN, making management of these lesions challenging. Biomarkers based on mutation
511 profiles may overcome this problem in the future (see biomarkers).

512 Clinical trial evidence regarding the benefits of screening high risk cohorts for pancreatic
513 cancer will be challenging to obtain due to the relatively low incidence of familial pancreatic cancer.
514 However, a mathematical model found that, depending on assumptions regarding the age of
515 screening and the characteristics of the screening test, screening could save lives in cohorts with a
516 risk of developing pancreatic cancer of at least 2.5 times that of the general population ¹⁶².

517 Since general screening is currently not available on a clinical level, primary prevention is
518 important: avoidance of smoking is the major practicable way for reducing the number of cases. In
519 developed nations tobacco control efforts are likely to result in future decreases in pancreatic cancer
520 incidence, but in developing countries incidence may rise due to increasing consumption of tobacco
521 products. Control of obesity is another possible preventive measure; thus integrated efforts to
522 address the international obesity epidemic may result in avoidance of a large number of pancreatic
523 cancer cases. Importantly though, known risk factors for pancreatic cancer are able to explain only
524 about a quarter to a third of the disease ⁴

525

526 **5. Management**

527 Palliative therapies, novel therapies and targets

528 Optimizing therapy for patients with pancreatic adenocarcinoma is a formidable challenge. The
529 majority of patients present with locally advanced and technically unresectable disease, usually
530 because of vascular involvement, or with widespread metastatic disease, generally to the liver and
531 peritoneum. Less than 20% of patients have resectable disease and, for those who undergo

532 resection, followed by adjuvant therapy, approximately 80% will relapse and ultimately die of their
533 disease.

534 For these reasons, discovering effective drug therapies for affected patients is of paramount
535 importance. Historically, very few active drugs have been identified. As described in
536 “microenvironment”, this disease is associated with profound desmoplastic stroma which, in
537 preclinical systems, has been shown to impede drug delivery ¹¹³. In addition, almost all cases are
538 driven by *KRAS* activation (see mutational landscape), a poor prognostic factor in cancer in general
539 and one that is often associated with treatment resistance. Nonetheless, some progress has been
540 made that is positively influencing the therapeutic landscape.

541 In 1997, gemcitabine, a nucleoside analogue, was approved by the Food and Drug
542 Administration (FDA) for therapy of pancreatic adenocarcinoma based on a randomized trial
543 comparing gemcitabine to bolus 5-fluorouracil (5FU) ¹⁶³. The primary endpoint was clinical benefit as
544 measured by symptom control and the survival advantage was quite modest. In 2005, erlotinib, a
545 tyrosine kinase inhibitor of EGFR, in combination with gemcitabine was FDA approved, again based
546 on a randomized trial that showed significant but minimal improvement over gemcitabine alone ¹⁶⁴.
547 However in 2011, a combination of 5FU, leucovorin, irinotecan and oxaliplatin (FOLFIRINOX) was
548 shown to have robust activity when compared to gemcitabine monotherapy ¹⁶⁵. This regimen,
549 however, has significant toxicities of diarrhea, nausea, fatigue, myelosuppression and neuropathy,
550 which are only partially controlled with anti-diarrheal and anti-emetic drugs. Thus the regimen is
551 usually reserved for patients 76 years of age or younger with an excellent performance status (ECOG
552 0-1). In 2012, the regimen of gemcitabine and albumin bound paclitaxel was introduced, again
553 showing improved efficacy ¹⁶⁶ in a randomized trial compared to gemcitabine alone, leading to FDA
554 approval of albumin bound paclitaxel for pancreatic cancer treatment. Toxicities with this therapy
555 are also significant: alopecia, myelosuppression, nausea, fatigue and neuropathy. However, this
556 regimen can be safely given to patients who are somewhat older or who have a slightly worse
557 performance status (ECOG 0-2)

558 The National Comprehensive Cancer Network (NCCN) guidelines list a number of other
559 combinations that have somewhat lower levels of evidence. These include pharmacokinetic
560 optimization of gemcitabine by fixed dose rate delivery ¹⁶⁷, gemcitabine and cisplatin (for patients
561 with germline DNA repair mutations) ¹⁶⁸, gemcitabine and capecitabine (based on a study showing
562 significant improvement in progression free survival) ¹⁶⁹, and fixed dose rate gemcitabine, taxotere
563 and capecitabine (GTX, an active regimen which has never been compared to gemcitabine
564 monotherapy) ¹⁷⁰.

565 The choice of second line therapy depends primarily on which regimen was used initially,
566 opting for a gemcitabine based regimen if FOLFIRINOX was initially used or a fluorinated pyrimidine

567 based regimen if gemcitabine and albumin bound paclitaxel was initially used. The FDA has recently
568 approved liposomal irinotecan to be used in combination with 5FU and leucovorin following first line
569 therapy with gemcitabine ¹⁷¹.

570 Locally advanced, unresectable disease (local extension usually around large vessels but
571 without obvious metastatic disease), has traditionally been treated with chemoradiation. However, a
572 pivotal trial of systemic therapy with gemcitabine monotherapy followed by chemoradiation or
573 continued chemotherapy, failed to show a survival benefit for the inclusion of radiation¹⁷². Whether
574 this is true following more active regimens awaits testing.

575 “Actionable” mutations are unusual in this disease but this may change as our knowledge
576 improves (see mutational landscape). Approximately 7% of patients have mutations involving DNA
577 repair genes (e.g. BRCA2, PALB2) raising hope that PARP inhibitors may be useful in this subset of
578 patients ³⁸.

579 Other innovative strategies currently in clinical trials include PEGPH20 (a pegylated
580 hyaluronidase for stromal modulation) ¹⁷³, ruxolitinib and momelotinib (JAK-STAT inhibitors) ¹⁷⁴,
581 ibrutinib (a BTK inhibitor) ¹⁷⁵, MM-141 (an IGF1R inhibitor), and palbociclib (a CDK4 and CDK6
582 inhibitor) among others. One of the emerging interests in pancreatic cancer therapy is
583 immunotherapy, since these tumors generally escape from immune surveillance through various
584 mechanisms such as low immunogenicity due to expression of PD-L1/2, the secretion of
585 immunosuppressive factors like TGF- β s, and an immunosuppressive microenvironment depleted of
586 effector T-cells ¹⁷⁶. Various strategies have been employed in this context such as vaccination
587 (CRS207 and GVAX engineered vaccines), to induce or reinforce pre-existing immune responses ¹⁷⁷,
588 immune checkpoint blockade to activate T cell function ^{119, 178}, or oncolytic virus therapy for induction
589 of inflammation, immunomodulation and cancer cell lysis ¹⁷⁹. Immunotherapies may be most
590 effective in cancers with microsatellite instability¹⁷⁷ as has been described in colorectal cancer ¹⁸⁰.
591 Given the great promise of immune-based therapies in other malignancies, there is reason to hope
592 that modulating the tumor microenvironment may also augment tumor control in pancreatic
593 adenocarcinoma.

594

595 Surgery

596 Surgery remains the only chance for cure of pancreatic cancer. Historically, surgical resection
597 of a pancreatic cancer was associated with unacceptably high morbidity and mortality rates ¹⁸¹,
598 especially in low volume centers ¹⁸². Further, the impact of surgery on the overall survival of
599 pancreatic cancer patients was questioned ^{183, 184}. Together, this had led to (surgical) therapeutic
600 nihilism and an underutilization of pancreatic cancer surgery ¹⁸⁵. Today, with increasing experience,
601 higher levels of centralization ^{186, 187}, and better perioperative management, pancreatic cancer

602 surgery can be carried out safely in high volume centers with acceptable mortality rates of less than
603 5%¹⁸⁸. Further, macroscopic complete tumor resection and adjuvant therapies results in robust 20%
604 5-year survival rates¹⁸⁹⁻¹⁹¹, with 40% to more than 50% in selected subgroups^{188, 192}.

605 For cancers of the pancreatic head, a partial pancreaticoduodenectomy (the so called
606 Whipple procedure) is usually required with or without partial resection of the distal stomach, with
607 both approaches not being relevantly different¹⁹³. For tumors of the tail a distal pancreatectomy
608 with splenectomy is carried out. A total pancreatectomy with resulting exocrine and endocrine
609 insufficiency (brittle diabetes) is rarely required except for tumors involving the entire length of the
610 organ, centrally located tumors, or for technical reasons¹⁹⁴. Pancreatic cancer surgery has
611 traditionally been carried out as an open procedure, but laparoscopic resections are increasingly
612 being performed. While there is a lack of high quality evidence, laparoscopic distal pancreatectomy is
613 considered a safe and effective option¹⁹⁵. In contrast, laparoscopic pancreaticoduodenectomy is a
614 demanding and complex procedure that is not considered standard at present¹⁹⁶, with increased
615 mortality being a potential issue in low volume hospitals¹⁹⁷. Robotic or computer-assisted resections
616 are feasible and safe in few specialized centers^{198, 199}. The long learning curve, training aspects and
617 high costs of this technique remain a problem.

618 At diagnosis, approximately 10-20% of the patients present with resectable tumors, 30-40%
619 with borderline resectable (BRPC) or locally advanced/unresectable tumors (LAPC), and 50-60% with
620 metastatic/systemic disease. Clear definitions of BRPC and LAPC had been lacking in the past, which
621 hampered the comparison of clinical trials and reliable outcome analysis. Today, classification of
622 resectability (BRPC, LAPC) depending on major arterial (celiac trunk, hepatic artery, superior
623 mesenteric artery) and venous (portal vein, superior mesenteric vein) vessel involvement follow the
624 most widely used guidelines of the National Comprehensive Cancer Network²⁰⁰, which have also
625 been adopted by the International Study Group for Pancreatic Surgery (ISGPS)²⁰¹ (figure 12).

626 With the aim to achieve complete tumor resection, technical aspects have been thoroughly
627 addressed in the past decade. Thus, pancreatic cancer resection including venous resection can be
628 performed safely with survival comparable to resection without venous involvement^{202, 203}. In
629 contrast, arterial resections are associated with increase morbidity and mortality and a disputable
630 survival benefit²⁰⁴, although in selected cases, special procedures might be indicated^{205, 206}.
631 Multivisceral resections also increase morbidity, but may be necessary to achieve complete tumor
632 resection with survival rates comparable to standard resections²⁰⁷. A standard lymphadenectomy
633 (ISGPS²⁰⁸) is advised, as more extended surgeries increase morbidity but do not incase survival^{209, 210}
634 (figure 13).

635 Resection of the primary tumor and distant metastasis is currently not indicated, since small
636 series did not show a consistent survival benefit^{211, 212}, although this seems to be different in a

637 subgroup of patients with lung metastasis, in whom resection might be a viable option to increase
638 survival ²¹³. Tumor debulking (R2 resection) is also currently not indicated since it substantially
639 increases morbidity/mortality without a survival benefit ²¹⁴. Microscopic clearance (R0 resections)
640 has been a matter of debate, because of different definitions and protocols ²¹⁵⁻²¹⁷. Due to the
641 anatomical location with close proximity to major intraabdominal vessels, R0 resections are rare, and
642 high rate of R1 resections reflect tumor growth patterns rather than surgical expertise ²¹⁵⁻²¹⁷.

643

644 Perioperative & adjuvant therapies

645 Since surgery alone is not sufficient to achieve long term survival ²¹⁸, adjuvant therapy is standard
646 following pancreatic cancer resection ^{189, 191}. Postoperative therapy with either gemcitabine ¹⁹¹ or 5FU
647 and leucovorin ¹⁸⁹ has demonstrated significant improvement on overall and 5 year survival
648 compared to observation. Many adjuvant studies have included radiation, usually chemoradiation, in
649 addition to a course of systemic chemotherapy ²¹⁹, but the true impact of the addition of radiation is
650 unknown. More active regimens such as FOLFIRINOX and gemcitabine and albumin bound paclitaxel
651 are now being studied in the adjuvant setting.

652 The success of neoadjuvant (preoperative) therapy in diseases such as breast and rectal
653 cancer and the availability of more active regimens, have stimulated interest in the application of
654 neoadjuvant treatment for pancreatic cancer. Data from high quality trials are missing; however,
655 reviewing the current evidence suggests that there is no benefit for neoadjuvant therapy for
656 resectable tumors (as compared to resection and adjuvant therapies), although the systemic therapy
657 used in these reports was of low effectiveness ²²⁰. Ongoing trials analyzing neoadjuvant therapy in
658 the setting of resectable tumors will help to answer this controversy ^{221, 222}. In contrast, evidence
659 suggests that approximately one third of initially BRPC/LAPC become resectable after neoadjuvant
660 therapy with similar outcomes as initially resectable tumors ²²⁰. Traditionally, most protocols have
661 used chemoradiation; however, the availability of more effective systemic therapies such as
662 FOLFIRINOX or albumin bound paclitaxel /Gemcitabine ^{165, 166}, has led to an increased interest in this
663 approach. First data using FOLFIRINOX in the neoadjuvant setting suggested a relevant rate of
664 conversion of BRPC/LAPC to resectable tumors ^{223, 224} and highlighted the difficulties in response
665 evaluation. Prospective randomized studies are needed for this approach as well as for the other
666 current options: gemcitabine-based chemotherapy, chemoradiation, or induction chemotherapy
667 followed by treatment with chemoradiation ^{225, 226}. Importantly, these patients should be discussed
668 by specialized multidisciplinary teams ²²⁷.

669

670

6. Quality of life

Quality of Life (QoL) assessment in pancreatic cancer is important because the disease is often incurable, and survival is short. The cancer commonly causes severe pain requiring opioid treatment, and may obstruct the bile duct causing jaundice, or the duodenum causing anorexia, nausea and vomiting. Besides these physical symptoms, patients with advanced incurable cancer face psychological and social distress, and spiritual challenges, which are summarized as end of life issues. Most of these patients experience unmet needs in several areas, most of them related to psychological or emotional distress and to medical communication or information regarding specific treatments and its potential benefits and risks ²²⁸.

The diagnosis is often late: approximately 30-40% have locally advanced disease and 50-60% have metastases at diagnosis ²²⁹, with median survival of only 6-9 months and 3 months respectively. The main aims of treatment are to relieve symptoms and to slow tumor progression. Even in the 10-20% who can undergo curative treatment, the effects of major pancreatic resection, chemotherapy or radiotherapy have a huge impact on QoL.

Two systems are available in multiple languages to assess QoL in patients with cancer, the EORTC QLQ-C30 ²²⁹ and the FACT ²³⁰. They assess general cancer symptoms and functional impairments, and have specific organ supplements. QoL issues identified by professionals and by patients as important are included in the EORTC QLQ-PAN26 (for pancreatic cancer) ²³¹ and the FACT(Hep) (hepatobiliary and pancreatic malignancies) ²³². Table 2 shows the issues covered by the EORTC system. Patients with pancreatic cancer may have impairment in any of these areas at some stage in their illness (see Box).

Long term survival is only possible after surgical resection (often with adjuvant therapy; see management). Surgery is a major physical and psychological challenge, with obvious immediate effects on QoL. One month after surgery, domains such as pain, fatigue, appetite, and bowel habit are worse than pre-operatively. Most domains recover to preoperative values or better at 3-6 months ^{233, 234}. Common problems after pancreatectomy are weight loss, abdominal pain, fatigue, foul stools, and diabetes mellitus but long term QoL is good in most domains ²³⁵.

The side effects of chemotherapy, including gastrointestinal toxicity, taste changes and fatigue may impair QoL, but effective therapy enhances QoL. In one randomized comparison of gemcitabine against PEFG (cisplatin, epirubicin, 5-fluorouracil, gemcitabine), emotional function, fatigue, overall QoL, pain, and flatulence all improved in both arms ²³⁶. Physical function, fatigue, appetite, and satisfaction with healthcare improved in 40–46% of partial responders compared with 0–12% of patients with stable disease.

704 The SCALOP trial ²³⁷ of palliative chemoradiation (CRT) confirmed the improvement in most
705 areas of QoL during induction chemotherapy. During CRT, fatigue, appetite loss, and gastrointestinal
706 symptoms were reported. However these all recovered within 3 weeks of finishing CRT.
707 QoL issues and the end of life situation of pancreatic cancer patients is often not adequately
708 considered in designing and reporting of clinical trials. For example, QoL issues or patient-reported
709 outcomes were considered in only 4 of 25 analyzed clinical trials for pancreatic cancer ²³⁸. In
710 conclusion, pancreatic cancer is a disease with major effects on QoL. Most treatments adversely
711 affect QoL. However, effective treatments lead to significant improvement of QoL, which often
712 justifies their use even in the palliative setting.

713

714 **Box - Quality of life issues**

715 Pain is a major issue for many patients; good pain control improves QoL.
716 Digestive disturbances arise from cancer effects on the gut and from toxicities of treatment: common
717 problems include loose stools, diarrhea and weight loss (from malabsorption) and constipation (from
718 taking opioids, and reduced appetite).

719

720

721

722 7. Outlook

723 Pancreatic cancer remains a deadly disease, yet the next decade holds promise to substantially
724 change the current dismal outlook for most pancreatic cancer patients. Key areas for basic and
725 clinical research are advances in our understanding of the molecular pathophysiology, precision
726 medicine ²³⁹ i.e., subgrouping pancreatic cancer patients for tailored therapies, and standardization
727 in clinical care and research.

728

729 Precision medicine

730 An in depth understanding of the molecular pathology of pancreatic cancer through large-scale -omic
731 and other approaches will be necessary for improved patient selection i.e., patient stratification to
732 specific therapies. As we understand more about the molecular pathology of pancreatic cancer, it is
733 becoming clear why trials of targeted therapies have mostly not been successful in the past, or were
734 associated with only incremental benefits. First, signaling in pancreatic cancer is complex, consisting
735 of multiple nodes and aberrant pathways. Thus, targeting single alterations is often ineffective due to
736 redundant signaling and complex cross-talks. Second, apart from four common mutations, which are
737 not currently druggable, essentially all of the currently druggable mutations in pancreatic cancer are
738 at low prevalence. As a consequence, any trial that did not select appropriate patients based on
739 molecular subtypes would not have detected a signal. At this prevalence, even if one appropriate
740 patient was captured by chance and had an exceptional response ²⁴⁰, it would constitute an anecdote
741 and not be informative.

742 To overcome these problems, druggable key signaling hubs, have to be identified. Second,
743 low prevalence alterations have to be targeted, possibly also in the context of grouping cancers with
744 common alterations rather than by the cell/organ or origin. Third, for effective treatment, new,
745 multipronged approaches need to be developed, which not only target cancer cells but also
746 reprogram the cancer stroma by modulating the interactions of stellate cells, endothelial cell,
747 immune cells and cancer cells. Fourth, for more effective therapy, pancreatic cancer has to be
748 detected at earlier stages, using novel biomarkers and multimodal imaging.

749 The wealth of data generated through large-scale approaches will aid in achieving these
750 goals, and the decreasing costs of whole genome sequencing ²⁴¹ will likely make this approach
751 routine in the coming decade. Clinical trials are mandatory to analyze different treatment options;
752 however, these trials require considerable effort from all involved stakeholders ²⁴² and with the
753 advent of more personalized medicine, clinical trials need increasingly to be carried out multicentric
754 and/or multinational settings.

755

756

757 Standardization in clinical care and research

758 Progress in the clinical management of pancreatic cancer patients had been hampered by the lack of
759 generally accepted definitions of clinical and outcome parameters. Unified definitions and their
760 routine use will be a challenge in the coming years. For example, definitions of resectability²⁰¹ and of
761 postoperative complications in general²⁴³ as well as for specific complications²⁴⁴⁻²⁴⁶ have been put
762 forward and are now widely accepted as are definitions of adverse events following chemo-, radio-,
763 or targeted therapy²⁴⁷, and of quality of life. This has made comparison of trials and cohort data
764 easier and greatly enhanced its clinical applicability. Indeed, it should be mandatory in the future to
765 publish data only if accepted definitions of clinical and outcome parameters were used. There are
766 still heterogeneous definitions in other areas, such as R1 definitions²¹⁷ and minimal datasets
767 required for clinical reporting have not been established, blocking progress in those areas. In the
768 upcoming years, it will be an important challenge to agree on the most important parameters.
769 Internationally accepted standards are a prerequisite, as are reduced institutional obstacles to
770 carrying out multicentric/multinational trials.

771 Further, the difficulties in the clinical management of pancreatic cancer are best highlighted
772 by pancreatic cancer surgery. There has to be a continuous effort to make pancreatic surgery safe.
773 While we fortunately left behind the period of mortality rates as high as 20-40%¹⁸¹, up to 5% (a
774 current quality mark) is still substantial. The key points (especially with the emergence of new
775 techniques) are patient selection, centralization and training. Centralization is an issue in many
776 countries^{186, 187}, and there has been a steady but slow progress in this area. It remains important to
777 treat pancreatic cancer patients in dedicated centers with transparent quality and outcome
778 reporting.

779

780 In summary, progress on different areas as highlighted in this primer will contribute towards a better
781 understanding of the complex molecular pathology, earlier diagnosis, and better, more individualized
782 patient care within the next decade.

783 **Authors' contribution**

784 Introduction (JK), Epidemiology (CLV), Risk factors (CLV, REN), Carcinogenesis, molecular biology, and
785 signaling (MK), Mutational landscape and subtypes (AVB), Microenvironment (MA), Metabolism
786 reprogramming (JK), Disease models (DAT), Biomarkers (MK), Diagnosis and pathology (RH),
787 Screening and Prevention (REN), Palliative therapies, novel therapies and targeted therapies (MAT),
788 Surgery (JK, JPN), Adjuvant therapy (MAT), Perioperative therapy (JK, MAT), Quality of life (CDJ),
789 Outlook (JK, AVB), Overview of the Primer (JK). All authors read and approved the final version of the
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791

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803

804

805 **Conflict of interest**

806 RHH receives royalty payments from Myriad Genetics for the PALB2 invention in a relationship that is
807 managed by Johns Hopkins University.

808 The following authors have no potential or actual conflicts of interest to disclose: AVB, CDJ, CLV, DAT,
809 JK, MAT, MK, REN

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814 **Tables**

815

816 Table 1: Genetic syndromes associated with pancreatic cancer (sorted by relative risk)

817

Syndrome	Gene	Relative Risk
Peutz Jeghers Syndrome	<i>LKB1/STK11</i> (Liver kinase B1 / serine threonine kinase 11)	132 ²⁴⁸
Hereditary pancreatitis	<i>PRSS1</i> (Protease serine 1)	53 ²⁴⁹ -67 ²⁵⁰
Familial atypical mole melanoma	<i>CDKN2A(p16)</i> (Cyclin-dependent kinase 2A)	22 ²⁵¹ -38 ²⁵²
Hereditary non-polyposis colon cancer (Lynch Syndrome)	<i>MSH2, MLH1, MSH6, PMS, PMS2</i> (mismatch repair genes)	9 ²⁵³
Familial pancreatic cancer (1 or more first degree relatives affected)	Unknown	9 (5-32 depending on number of relatives affected) ¹⁵⁸
Cystic fibrosis	<i>CFTR</i> (Cystic fibrosis transmembrane conductance regulator)	5 ²⁵⁴
<i>Familial adenomatous polyposis*</i>	<i>APC (adenomatous polyposis coli)</i>	5 ²⁵⁵
Breast and ovarian cancer syndrome	<i>BRCA1/BRCA2/PALB2</i> (Breast cancer susceptibility genes 1 and 2)	2 ²⁵⁶ -4 ²⁵⁷
Li Fraumeni syndrome	<i>TP53</i> (tumor protein p53)	Unknown ²⁵⁸

818

819 **associated with ampullary cancer and duodenal cancer, not with pancreatic ductal adenocarcinoma*

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821

822 Table 2: Scales and items of the EORTC QLQ-C30 and EORTC QLQ-PAN26. Scales contain a number of
823 related items, which are reported as one value per scale. Both questionnaires contain some single
824 items, which are each reported separately. The 4 groups of single items shown for the PAN26 are not
825 reported as scales.

QLQ-C30		QLQ-PAN26	
	Number of items		Number of items
<i>Global health and QoL status</i>	2		
Function Scales			
Physical functioning	5	Body image	2
Role functioning	2	Satisfaction with health care	2
Emotional functioning	4	Sexuality	2
Cognitive functioning	2		
Social functioning	2		
Symptom Scales			
Fatigue	3	Pancreatic pain	4
Nausea and vomiting	2	Digestive symptoms	2
Pain	2	Altered bowel habit	2
		Hepatic	2
Single Items			
Insomnia	1	Emotional	2
Appetite loss	1	Side effects	3
Constipation	1	Cachexia	2
Diarrhoea	1	Other GI symptoms	3
Financial difficulties	1		

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829

830 **Figure legends**

831

832 Figure 1:

833 Estimated age-standardized mortality (A) and incidence (B) rates for both sexes (per 100,000
834 persons) 2012 ¹.

835

836 Figure 2:

837 Example of an aberrant signaling node in pancreatic cancer. c-Met, integrin- β 1 and other integrins,
838 neuropilin 1 (NP1), and CD44 (notably, CD44v6) are often overexpressed in pancreatic cancer and
839 may physically associate into a complex. Moreover, EGFR HER2 and HER3 are also overexpressed in
840 PDAC, and EGFR heterodimerizes with c-Met, HER2 and HER3. The cross-talk between EGFR family
841 members and between EGFR-c-Met, occurs in the context of mutated *KRAS* and high levels of HGF
842 and ligands that bind to the EGFR family, leading to marked activation of p-STAT3 and p-AKT pro-
843 survival signaling, and enhanced mitogenesis, invasion metastasis and chemoresistance. Moreover,
844 aberrant TGF- β signaling through the TGF- β receptor heterotetramer complex leads to activation of
845 non-canonical pathways that promote proliferation and invasion, and to upregulation of Wnt7b
846 which activates frizzled receptors (FZD) to induce mitogenesis. Thus, loss of TGF- β -dependent growth
847 inhibition is associated with the conversion of TGF- β to a direct pancreatic cancer cell mitogen.

848

849 Figure 3:

850 The normal pancreas. The normal pancreas consists of enzyme secreting acinar cell, bicarbonate
851 secreting ductal cells, centro-acinar cells that are the geographic transition between acinar and
852 ductal cells, hormone-secreting endocrine islets and relatively inactive stellate cells.

853

854 Figure 4:

855 Commonly mutated genes that characterize pancreatic cancer determined using MutSig analysis (A).
856 Subtyping of pancreatic cancer using structural rearrangements (B) ³⁸.

857

858 Figure 5:

859 The Figure depicts the known interactions of pancreatic stellate cells with i) cancer cells ii) other cells
860 in the desmoplastic reaction including immune cells, endothelial cells and neurons; and iii) beta cells
861 in islets. All of these interactions are thought to facilitate the progression of pancreatic cancer.

862

863 Figure 6:

864 The figure depicts schematically the metabolic reprogramming in pancreatic cancer.

865

866 Figure 7:

867 Models of pancreatic cancer. Genetically engineered mouse models, and organoid models prepared
868 from GEMMs and patients are available experimental systems for pancreatic cancer research
869 laboratories.

870

871 Figure 8:

872 Schematic representation of types of biomarkers for the diagnosis of pancreatic cancer. Circulating
873 tumor cells (CTC) are unlikely to pick up early pre-metastatic pancreatic cancer. CA19-9 is not
874 sufficiently sensitive or specific to be useful for early pancreatic cancer diagnosis. Branched chain
875 amino acids (BCAA) and circulating tumor DNA (ctDNA) are elevated in early stages in both human
876 and murine pancreatic cancer, and could help with early diagnosis. MicroRNAs (miR) and proteins
877 can be detected in the plasma or within exosomes.

878

879 Figure 9:

880 Multidetector computed tomography scan of the abdomen demonstrating a hypodense pancreatic
881 lesion (red arrowheads) obstruction/infiltrating the superior mesenteric vein (blue arrowheads. Note
882 the visible dilated venous collaterals proximal of the obstruction (blue arrows). Kindly provided by
883 Konstantin Holzapfel, Technical University of Munich, Germany.

884

885 Figure 10:

886 Magnetic resonance cholangiopancreatography demonstrating a dilated bile duct (green
887 arrowheads) and pancreatic duct (yellow arrowheads) "double duct sign" due to a suspected
888 pancreatic head lesion (red dotted circle). Kindly provided by Konstantin Holzapfel, Technical
889 University of Munich, Germany.

890

891 Figure 11:

892 Infiltrating ductal adenocarcinoma of the pancreas. Note the extensive fibrosis associated with the
893 neoplastic glands (center). (Hematoxylin and eosin stained)

894

895 Figure 12:

896 Anatomical location of a pancreatic cancer (red arrows) of the pancreatic head/body in close
897 proximity to the celiac trunk and the superior mesenteric vein/portal vein. Kindly provided by
898 Matthias Maak, University of Erlangen, Germany.

899

900 Figure 13:
901 Intraoperative situs following resection of a pancreatic head cancer with standard lymphadenectomy
902 ²⁰⁸. The portal vein (blue vessel loop) and the common hepatic artery (white vessel loop) are
903 indicated. The gastroduodenal artery is clip ligated (white arrow). * cut end of the pancreas with the
904 pancreatic duct (black arrow).
905
906

References

1. GLOBOCAN 2012. Ferlay J, Soerjomataram I, Ervik M, Dikshit R, Eser S, Mathers C, Rebelo M, Parkin DM, Forman D, Bray, F. GLOBOCAN 2012 v1.0, Cancer Incidence and Mortality Worldwide: IARC CancerBase No. 11 [Internet]. Lyon, France: International Agency for Research on Cancer. Available from <http://globocan.iarc.fr>. Accessed 12 Dec 2013. (2013).
2. Rahib, L. et al. Projecting cancer incidence and deaths to 2030: the unexpected burden of thyroid, liver, and pancreas cancers in the United States. *Cancer Res* **74**, 2913-21 (2014).
3. Siegel, R.L., Miller, K.D. & Jemal, A. Cancer statistics, 2015. *CA Cancer J Clin* **65**, 5-29 (2015).
4. Rosato, V. et al. Population attributable risk for pancreatic cancer in Northern Italy. *Pancreas* **44**, 216-20 (2015).
5. Boffetta, P., Boccia S & La Vecchia C. A Quick Guide to Cancer Epidemiology. *Springer book Series: SpringerBriefs in Cancer Research IX*, 101 p. Springer Cham Heidelberg New York Dordrecht London (2014).
6. CI5-X. Cancer Incidence in Five Continents, Vol. X (electronic version) Lyon, IARC. <http://ci5.iarc.fr> last accessed on [date]. . (2013).
7. Bosetti, C. et al. Pancreatic cancer: overview of descriptive epidemiology. *Mol Carcinog* **51**, 3-13 (2012).
8. Malvezzi, M. et al. European cancer mortality predictions for the year 2015: does lung cancer have the highest death rate in EU women? *Ann Oncol* **26**, 779-86 (2015).
9. Anderson, K., Mack, T. & Silverman, D. Cancer of the Pancreas. In: Schottenfeld D, Fraumeni JF (eds) *Cancer Epidemiology and Prevention*. Oxford University Press, New York. pp 721-762. (2006).
10. Bosetti, C. et al. Cigarette smoking and pancreatic cancer: an analysis from the International Pancreatic Cancer Case-Control Consortium (PanC4). *Ann Oncol* **23**, 1880-8 (2012).
11. Parkin, D.M. 2. Tobacco-attributable cancer burden in the UK in 2010. *Br J Cancer* **105 Suppl 2**, S6-S13 (2011).
12. Whitman, D.C. et al. Cancers in Australia in 2010 attributable to modifiable factors: introduction and overview. *Aust N Z J Public Health* **39**, 403-7 (2015).
13. Boffetta, P., Hecht, S., Gray, N., Gupta, P. & Straif, K. Smokeless tobacco and cancer. *Lancet Oncol* **9**, 667-75 (2008).
14. Behrens, G. et al. Physical activity and risk of pancreatic cancer: a systematic review and meta-analysis. *Eur J Epidemiol* **30**, 279-98 (2015).
15. Genkinger, J.M. et al. Central adiposity, obesity during early adulthood, and pancreatic cancer mortality in a pooled analysis of cohort studies. *Ann Oncol* **26**, 2257-66 (2015).
16. Larsson, S.C. & Wolk, A. Red and processed meat consumption and risk of pancreatic cancer: meta-analysis of prospective studies. *Br J Cancer* **106**, 603-7 (2012).
17. Rohrmann, S. et al. Meat and fish consumption and risk of pancreatic cancer: results from the European Prospective Investigation into Cancer and Nutrition. *Int J Cancer* **132**, 617-24 (2013).
18. Bosetti, C. et al. Nutrient-based dietary patterns and pancreatic cancer risk. *Ann Epidemiol* **23**, 124-8 (2013).
19. Tramacere, I. et al. Alcohol drinking and pancreatic cancer risk: a meta-analysis of the dose-risk relation. *Int J Cancer* **126**, 1474-86 (2010).
20. Lucenteforte, E. et al. Alcohol consumption and pancreatic cancer: a pooled analysis in the International Pancreatic Cancer Case-Control Consortium (PanC4). *Ann Oncol* **23**, 374-82 (2012).
21. Bosetti, C. et al. Diabetes, antidiabetic medications, and pancreatic cancer risk: an analysis from the International Pancreatic Cancer Case-Control Consortium. *Ann Oncol* **25**, 2065-72 (2014).

- 957 22. Chari, S.T. et al. Pancreatic cancer-associated diabetes mellitus: prevalence and temporal
958 association with diagnosis of cancer. *Gastroenterology* **134**, 95-101 (2008).
- 959 23. Bosetti, C. et al. Ulcer, gastric surgery and pancreatic cancer risk: an analysis from the
960 International Pancreatic Cancer Case-Control Consortium (PanC4). *Ann Oncol* **24**, 2903-10
961 (2013).
- 962 24. Turati, F. et al. Family history of cancer and the risk of cancer: a network of case-control
963 studies. *Ann Oncol* **24**, 2651-6 (2013).
- 964 25. Rizzato, C. et al. ABO blood groups and pancreatic cancer risk and survival: results from the
965 PANcreatic Disease ReseArch (PANDORA) consortium. *Oncol Rep* **29**, 1637-44 (2013).
- 966 26. Hezel, A.F., Kimmelman, A.C., Stanger, B.Z., Bardeesy, N. & Depinho, R.A. Genetics and
967 biology of pancreatic ductal adenocarcinoma. *Genes Dev* **20**, 1218-49 (2006).
- 968 27. Preis, M. & Korc, M. Signaling pathways in pancreatic cancer. *Crit Rev Eukaryot Gene Expr* **21**,
969 115-29 (2011).
- 970 28. Pasca di Magliano, M. et al. Common activation of canonical Wnt signaling in pancreatic
971 adenocarcinoma. *PLoS One* **2**, e1155 (2007).
- 972 29. Bublil, E.M. & Yarden, Y. The EGF receptor family: spearheading a merger of signaling and
973 therapeutics. *Curr Opin Cell Biol* **19**, 124-34 (2007).
- 974 30. Matsushita, A., Gotze, T. & Korc, M. Hepatocyte growth factor-mediated cell invasion in
975 pancreatic cancer cells is dependent on neuropilin-1. *Cancer Res* **67**, 10309-16 (2007).
- 976 31. Morello, V. et al. beta1 integrin controls EGFR signaling and tumorigenic properties of lung
977 cancer cells. *Oncogene* **30**, 4087-96 (2011).
- 978 32. Tanizaki, J., Okamoto, I., Sakai, K. & Nakagawa, K. Differential roles of trans-phosphorylated
979 EGFR, HER2, HER3, and RET as heterodimerisation partners of MET in lung cancer with MET
980 amplification. *Br J Cancer* **105**, 807-13 (2011).
- 981 33. Gore, A.J., Deitz, S.L., Palam, L.R., Craven, K.E. & Korc, M. Pancreatic cancer-associated
982 retinoblastoma 1 dysfunction enables TGF-beta to promote proliferation. *J Clin Invest* **124**,
983 338-52 (2014).
- 984 34. Kleeff, J. et al. The cell-surface heparan sulfate proteoglycan glypican-1 regulates growth
985 factor action in pancreatic carcinoma cells and is overexpressed in human pancreatic cancer.
986 *J Clin Invest* **102**, 1662-73 (1998).
- 987 35. Jones, S. et al. Core signaling pathways in human pancreatic cancers revealed by global
988 genomic analyses. *Science* **321**, 1801-6 (2008).
- 989 36. Wang, L. et al. Whole-exome sequencing of human pancreatic cancers and characterization
990 of genomic instability caused by MLH1 haploinsufficiency and complete deficiency. *Genome*
991 *Res* **22**, 208-19 (2012).
- 992 37. Biankin, A.V. et al. Pancreatic cancer genomes reveal aberrations in axon guidance pathway
993 genes. *Nature* **491**, 399-405 (2012).
- 994 38. Waddell, N. et al. Whole genomes redefine the mutational landscape of pancreatic cancer.
995 *Nature* **518**, 495-501 (2015).
- 996 39. Witkiewicz, A.K. et al. Whole-exome sequencing of pancreatic cancer defines genetic
997 diversity and therapeutic targets. *Nat Commun* **6**, 6744 (2015).
- 998 40. Chang, D.K., Grimmond, S.M. & Biankin, A.V. Pancreatic Cancer Genomics. *Current Opinion in*
999 *Genetics & Development*. **24**, 74-81 (2014).
- 1000 41. Alexandrov, L.B. et al. Signatures of mutational processes in human cancer. *Nature* **500**, 415-
1001 21 (2013).
- 1002 42. Blackford, A. et al. Genetic mutations associated with cigarette smoking in pancreatic cancer.
1003 *Cancer Res* **69**, 3681-8 (2009).
- 1004 43. Nones, K. et al. Genome-wide DNA methylation patterns in pancreatic ductal
1005 adenocarcinoma reveal epigenetic deregulation of SLIT-ROBO, ITGA2 and MET signaling. *Int J*
1006 *Cancer* (2014).
- 1007 44. Collisson, E.A. et al. Subtypes of pancreatic ductal adenocarcinoma and their differing
1008 responses to therapy. *Nature medicine* **17**, 500-503 (2011).

- 1009 45. Moffitt, R.A. et al. Virtual microdissection identifies distinct tumor- and stroma-specific
1010 subtypes of pancreatic ductal adenocarcinoma. *Nat Genet* (2015).
- 1011 46. Gore, J. & Korc, M. Pancreatic cancer stroma: friend or foe? *Cancer Cell* **25**, 711-2 (2014).
- 1012 47. Biankin, A.V. & Maitra, A. Subtyping Pancreatic Cancer. *Cancer Cell* (2015).
- 1013 48. Kottakis, F. & Bardeesy, N. Gene signatures from pancreatic cancer tumor and stromal cells
1014 predict disease outcome. *Nature Gen* **47**, 1102-3 (2015).
- 1015 49. Seymour, A.B. et al. Allelotype of pancreatic adenocarcinoma. *Cancer Res* **54**, 2761-4 (1994).
- 1016 50. Clark, C.E. et al. Dynamics of the immune reaction to pancreatic cancer from inception to
1017 invasion. *Cancer Res* **67**, 9518-27 (2007).
- 1018 51. Amedei, A., Niccolai, E. & Prisco, D. Pancreatic cancer: role of the immune system in cancer
1019 progression and vaccine-based immunotherapy. *Hum Vaccin Immunother* **10**, 3354-68
1020 (2014).
- 1021 52. Inman, K.S., Francis, A.A. & Murray, N.R. Complex role for the immune system in initiation
1022 and progression of pancreatic cancer. *World J Gastroenterol* **20**, 11160-81 (2014).
- 1023 53. Lutz, E.R. et al. Immunotherapy converts nonimmunogenic pancreatic tumors into
1024 immunogenic foci of immune regulation. *Cancer Immunol Res* **2**, 616-31 (2014).
- 1025 54. Erkan, M. et al. The activated stroma index is a novel and independent prognostic marker in
1026 pancreatic ductal adenocarcinoma. *Clin Gastroenterol Hepatol* **6**, 1155-61 (2008).
- 1027 55. Bever, K.M. et al. The prognostic value of stroma in pancreatic cancer in patients receiving
1028 adjuvant therapy. *HPB (Oxford)* (2014).
- 1029 56. Ozdemir, B.C. et al. Depletion of carcinoma-associated fibroblasts and fibrosis induces
1030 immunosuppression and accelerates pancreas cancer with reduced survival. *Cancer Cell* **25**,
1031 719-34 (2014).
- 1032 57. Apte, M.V., Wilson, J.S., Lugea, A. & Pandol, S.J. A starring role for stellate cells in the
1033 pancreatic cancer microenvironment. *Gastroenterology* **144**, 1210-9 (2013).
- 1034 58. Hamada, S. et al. Pancreatic stellate cells enhance stem cell-like phenotypes in pancreatic
1035 cancer cells. *Biochem Biophys Res Commun* **421**, 349-54 (2012).
- 1036 59. Xu, Z. et al. Role of pancreatic stellate cells in pancreatic cancer metastasis. *Am J Pathol* **177**,
1037 2585-96 (2010).
- 1038 60. Patel, M.B. et al. The role of the hepatocyte growth factor/c-MET pathway in pancreatic
1039 stellate cell-endothelial cell interactions: antiangiogenic implications in pancreatic cancer.
1040 *Carcinogenesis* (2014).
- 1041 61. Ene-Obong, A. et al. Activated Pancreatic Stellate Cells Sequester CD8+ T-Cells to Reduce
1042 Their Infiltration of the Juxtatumoral Compartment of Pancreatic Ductal Adenocarcinoma.
1043 *Gastroenterology* **45**, 1121-1132 (2013).
- 1044 62. Tang, D. et al. High expression of Galectin-1 in pancreatic stellate cells plays a role in the
1045 development and maintenance of an immunosuppressive microenvironment in pancreatic
1046 cancer. *Int J Cancer* **130**, 2337-48 (2012).
- 1047 63. Mace, T.A., Bloomston, M. & Lesinski, G.B. Pancreatic cancer-associated stellate cells: A
1048 viable target for reducing immunosuppression in the tumor microenvironment.
1049 *Oncoimmunology* **2**, e24891 (2013).
- 1050 64. Ma, Y., Hwang, R.F., Logsdon, C.D. & Ullrich, S.E. Dynamic mast cell-stromal cell interactions
1051 promote growth of pancreatic cancer. *Cancer Res* **73**, 3927-37 (2013).
- 1052 65. Shi, C. et al. Fibrogenesis in pancreatic cancer is a dynamic process regulated by macrophage-
1053 stellate cell interaction. *Lab Invest* **94**, 409-21 (2014).
- 1054 66. Li, X. et al. Sonic hedgehog paracrine signaling activates stromal cells to promote perineural
1055 invasion in pancreatic cancer. *Clin Cancer Res* **20**, 4326-38 (2014).
- 1056 67. Kikuta, K. et al. Pancreatic stellate cells reduce insulin expression and induce apoptosis in
1057 pancreatic beta-cells. *Biochem Biophys Res Commun* **433**, 292-7 (2013).
- 1058 68. Rhim, A.D. et al. Stromal elements act to restrain, rather than support, pancreatic ductal
1059 adenocarcinoma. *Cancer Cell* **25**, 735-47 (2014).

1060 69. Erkan, M. et al. Cancer-stellate cell interactions perpetuate the hypoxia-fibrosis cycle in
1061 pancreatic ductal adenocarcinoma. *Neoplasia* **11**, 497-508 (2009).

1062 70. Kong, B. et al. Hypoxia-induced endoplasmic reticulum stress characterizes a necrotic
1063 phenotype of pancreatic cancer. *Oncotarget* (2015).

1064 71. Schonenberger, M.J. & Kovacs, W.J. Hypoxia signaling pathways: modulators of oxygen-
1065 related organelles. *Front Cell Dev Biol* **3**, 42 (2015).

1066 72. Ye, L.Y. et al. Hypoxia-inducible factor 1alpha expression and its clinical significance in
1067 pancreatic cancer: a meta-analysis. *Pancreatology* **14**, 391-7 (2014).

1068 73. Doherty, J.R. & Cleveland, J.L. Targeting lactate metabolism for cancer therapeutics. *J Clin*
1069 *Invest* **123**, 3685-92 (2013).

1070 74. Cohen, R. et al. Targeting cancer cell metabolism in pancreatic adenocarcinoma. *Oncotarget*
1071 **6**, 16832-47 (2015).

1072 75. Ying, H. et al. Oncogenic Kras maintains pancreatic tumors through regulation of anabolic
1073 glucose metabolism. *Cell* **149**, 656-70 (2012).

1074 76. Son, J. et al. Glutamine supports pancreatic cancer growth through a KRAS-regulated
1075 metabolic pathway. *Nature* **496**, 101-5 (2013).

1076 77. Ma, Z., Vocadlo, D.J. & Vosseller, K. Hyper-O-GlcNAcylation is anti-apoptotic and maintains
1077 constitutive NF-kappaB activity in pancreatic cancer cells. *J Biol Chem* **288**, 15121-30 (2013).

1078 78. Metallo, C.M. et al. Reductive glutamine metabolism by IDH1 mediates lipogenesis under
1079 hypoxia. *Nature* **481**, 380-4 (2012).

1080 79. Wise, D.R. et al. Hypoxia promotes isocitrate dehydrogenase-dependent carboxylation of
1081 alpha-ketoglutarate to citrate to support cell growth and viability. *Proc Natl Acad Sci U S A*
1082 **108**, 19611-6 (2011).

1083 80. Walter, K. et al. Serum fatty acid synthase as a marker of pancreatic neoplasia. *Cancer*
1084 *Epidemiol Biomarkers Prev* **18**, 2380-5 (2009).

1085 81. Yang, S. et al. Pancreatic cancers require autophagy for tumor growth. *Genes Dev* **25**, 717-29
1086 (2011).

1087 82. Commisso, C. et al. Macropinocytosis of protein is an amino acid supply route in Ras-
1088 transformed cells. *Nature* **497**, 633-7 (2013).

1089 83. Caldas, C. et al. Frequent somatic mutations and homozygous deletions of the p16 (MTS1)
1090 gene in pancreatic adenocarcinoma. *Nat Genet* **8**, 27-32 (1994).

1091 84. Hingorani, S.R. et al. Preinvasive and invasive ductal pancreatic cancer and its early detection
1092 in the mouse. *Cancer Cell* **4**, 437-50 (2003).

1093 85. Guerra, C. et al. Chronic pancreatitis is essential for induction of pancreatic ductal
1094 adenocarcinoma by K-Ras oncogenes in adult mice. *Cancer Cell* **11**, 291-302 (2007).

1095 86. Aguirre, A.J. et al. Activated Kras and Ink4a/Arf deficiency cooperate to produce metastatic
1096 pancreatic ductal adenocarcinoma. *Genes Dev* **17**, 3112-26 (2003).

1097 87. Huang, L. et al. Ductal pancreatic cancer modeling and drug screening using human
1098 pluripotent stem cell- and patient-derived tumor organoids. *Nat Med* (2015).

1099 88. Boj, S.F. et al. Organoid models of human and mouse ductal pancreatic cancer. *Cell* **160**, 324-
1100 38 (2015).

1101 89. Huch, M. et al. Unlimited in vitro expansion of adult bi-potent pancreas progenitors through
1102 the Lgr5/R-spondin axis. *EMBO J* **32**, 2708-21 (2013).

1103 90. Bardeesy, N. et al. Smad4 is dispensable for normal pancreas development yet critical in
1104 progression and tumor biology of pancreas cancer. *Genes Dev* **20**, 3130-46 (2006).

1105 91. Hingorani, S.R. et al. Trp53R172H and KrasG12D cooperate to promote chromosomal
1106 instability and widely metastatic pancreatic ductal adenocarcinoma in mice. *Cancer Cell* **7**,
1107 469-83 (2005).

1108 92. Hruban, R.H. et al. Pathology of genetically engineered mouse models of pancreatic exocrine
1109 cancer: consensus report and recommendations. *Cancer Res* **66**, 95-106 (2006).

1110 93. Izeradjene, K. et al. Kras(G12D) and Smad4/Dpc4 haploinsufficiency cooperate to induce
1111 mucinous cystic neoplasms and invasive adenocarcinoma of the pancreas. *Cancer Cell* **11**,
1112 229-43 (2007).

1113 94. Ijichi, H. et al. Aggressive pancreatic ductal adenocarcinoma in mice caused by pancreas-
1114 specific blockade of transforming growth factor-beta signaling in cooperation with active
1115 Kras expression. *Genes Dev* **20**, 3147-60 (2006).

1116 95. Mann, K.M. et al. Sleeping Beauty mutagenesis reveals cooperating mutations and pathways
1117 in pancreatic adenocarcinoma. *Proc Natl Acad Sci U S A* **109**, 5934-41 (2012).

1118 96. Perez-Mancera, P.A. et al. The deubiquitinase USP9X suppresses pancreatic ductal
1119 adenocarcinoma. *Nature* **486**, 266-70 (2012).

1120 97. Rad, R. et al. A conditional piggyBac transposition system for genetic screening in mice
1121 identifies oncogenic networks in pancreatic cancer. *Nat Genet* **47**, 47-56 (2015).

1122 98. Navas, C. et al. EGF receptor signaling is essential for k-ras oncogene-driven pancreatic ductal
1123 adenocarcinoma. *Cancer Cell* **22**, 318-30 (2012).

1124 99. Eser, S. et al. Selective requirement of PI3K/PDK1 signaling for Kras oncogene-driven
1125 pancreatic cell plasticity and cancer. *Cancer Cell* **23**, 406-20 (2013).

1126 100. DeNicola, G.M. et al. Oncogene-induced Nrf2 transcription promotes ROS detoxification and
1127 tumorigenesis. *Nature* **475**, 106-9 (2011).

1128 101. Collins, M.A. et al. Oncogenic Kras is required for both the initiation and maintenance of
1129 pancreatic cancer in mice. *J Clin Invest* **122**, 639-53 (2012).

1130 102. Saborowski, M. et al. A modular and flexible ESC-based mouse model of pancreatic cancer.
1131 *Genes Dev* **28**, 85-97 (2014).

1132 103. Kopp, J.L. et al. Identification of Sox9-dependent acinar-to-ductal reprogramming as the
1133 principal mechanism for initiation of pancreatic ductal adenocarcinoma. *Cancer Cell* **22**, 737-
1134 50 (2012).

1135 104. von Figura, G. et al. The chromatin regulator Brg1 suppresses formation of intraductal
1136 papillary mucinous neoplasm and pancreatic ductal adenocarcinoma. *Nat Cell Biol* **16**, 255-67
1137 (2014).

1138 105. Gidekel Friedlander, S.Y. et al. Context-dependent transformation of adult pancreatic cells by
1139 oncogenic K-Ras. *Cancer Cell* **16**, 379-89 (2009).

1140 106. Habbe, N. et al. Spontaneous induction of murine pancreatic intraepithelial neoplasia
1141 (mPanIN) by acinar cell targeting of oncogenic Kras in adult mice. *Proc Natl Acad Sci U S A*
1142 **105**, 18913-8 (2008).

1143 107. Pylayeva-Gupta, Y., Lee, K.E., Hajdu, C.H., Miller, G. & Bar-Sagi, D. Oncogenic Kras-induced
1144 GM-CSF production promotes the development of pancreatic neoplasia. *Cancer Cell* **21**, 836-
1145 47 (2012).

1146 108. Bayne, L.J. et al. Tumor-derived granulocyte-macrophage colony-stimulating factor regulates
1147 myeloid inflammation and T cell immunity in pancreatic cancer. *Cancer Cell* **21**, 822-35
1148 (2012).

1149 109. Gilbert, M. et al. Pancreatic cancer-induced cachexia is Jak2-dependent in mice. *J Cell*
1150 *Physiol* **229**, 1437-43 (2014).

1151 110. Stopczynski, R.E. et al. Neuroplastic changes occur early in the development of pancreatic
1152 ductal adenocarcinoma. *Cancer Res* **74**, 1718-27 (2014).

1153 111. Dawson, D.W. et al. High-fat, high-calorie diet promotes early pancreatic neoplasia in the
1154 conditional KrasG12D mouse model. *Cancer Prev Res (Phila)* **6**, 1064-73 (2013).

1155 112. Zambirinis, C.P. et al. TLR9 ligation in pancreatic stellate cells promotes tumorigenesis. *J Exp*
1156 *Med* (2015).

1157 113. Olive, K.P. et al. Inhibition of Hedgehog signaling enhances delivery of chemotherapy in a
1158 mouse model of pancreatic cancer. *Science* **324**, 1457-61 (2009).

1159 114. Kim, E.J. et al. Pilot clinical trial of hedgehog pathway inhibitor GDC-0449 (vismodegib) in
1160 combination with gemcitabine in patients with metastatic pancreatic adenocarcinoma. *Clin*
1161 *Cancer Res* **20**, 5937-45 (2014).

- 1162 115. Catenacci, D.V. et al. Randomized Phase Ib/II Study of Gemcitabine Plus Placebo or
1163 Vismodegib, a Hedgehog Pathway Inhibitor, in Patients With Metastatic Pancreatic Cancer. *J*
1164 *Clin Oncol* (2015).
- 1165 116. Provenzano, P.P. et al. Enzymatic targeting of the stroma ablates physical barriers to
1166 treatment of pancreatic ductal adenocarcinoma. *Cancer Cell* **21**, 418-29 (2012).
- 1167 117. Jacobetz, M.A. et al. Hyaluronan impairs vascular function and drug delivery in a mouse
1168 model of pancreatic cancer. *Gut* **62**, 112-20 (2013).
- 1169 118. Sherman, M.H. et al. Vitamin D receptor-mediated stromal reprogramming suppresses
1170 pancreatitis and enhances pancreatic cancer therapy. *Cell* **159**, 80-93 (2014).
- 1171 119. Feig, C. et al. Targeting CXCL12 from FAP-expressing carcinoma-associated fibroblasts
1172 synergizes with anti-PD-L1 immunotherapy in pancreatic cancer. *Proc Natl Acad Sci U S A*
1173 **110**, 20212-7 (2013).
- 1174 120. Beatty, G.L. et al. CD40 agonists alter tumor stroma and show efficacy against pancreatic
1175 carcinoma in mice and humans. *Science* **331**, 1612-6 (2011).
- 1176 121. Beatty, G.L. et al. Exclusion of T Cells From Pancreatic Carcinomas in Mice Is Regulated by
1177 Ly6C(low) F4/80(+) Extratumoral Macrophages. *Gastroenterology* **149**, 201-10 (2015).
- 1178 122. Keenan, B.P. et al. A Listeria vaccine and depletion of T-regulatory cells activate immunity
1179 against early stage pancreatic intraepithelial neoplasms and prolong survival of mice.
1180 *Gastroenterology* **146**, 1784-94 e6 (2014).
- 1181 123. Alagesan, B. et al. Combined MEK and PI3K inhibition in a mouse model of pancreatic cancer.
1182 *Clin Cancer Res* **21**, 396-404 (2015).
- 1183 124. Cook, N. et al. Gamma secretase inhibition promotes hypoxic necrosis in mouse pancreatic
1184 ductal adenocarcinoma. *J Exp Med* **209**, 437-44 (2012).
- 1185 125. Bausch, D. et al. Plectin-1 as a novel biomarker for pancreatic cancer. *Clin Cancer Res* **17**, 302-
1186 9 (2011).
- 1187 126. Neesse, A. et al. Claudin-4-targeted optical imaging detects pancreatic cancer and its
1188 precursor lesions. *Gut* **62**, 1034-43 (2013).
- 1189 127. Faca, V.M. et al. A mouse to human search for plasma proteome changes associated with
1190 pancreatic tumor development. *PLoS Med* **5**, e123 (2008).
- 1191 128. Melo, S.A. et al. Glypican-1 identifies cancer exosomes and detects early pancreatic cancer.
1192 *Nature* **523**, 177-82 (2015).
- 1193 129. Chiou, S.H. et al. Pancreatic cancer modeling using retrograde viral vector delivery and in vivo
1194 CRISPR/Cas9-mediated somatic genome editing. *Genes Dev* **29**, 1576-85 (2015).
- 1195 130. Yachida, S. et al. Distant metastasis occurs late during the genetic evolution of pancreatic
1196 cancer. *Nature* **467**, 1114-7 (2010).
- 1197 131. Lennon, A.M. et al. The early detection of pancreatic cancer: what will it take to diagnose and
1198 treat curable pancreatic neoplasia? *Cancer Res* **74**, 3381-9 (2014).
- 1199 132. Sausen, M. et al. Clinical implications of genomic alterations in the tumour and circulation of
1200 pancreatic cancer patients. *Nat Commun* **6**, 7686 (2015).
- 1201 133. Mayers, J.R. et al. Elevation of circulating branched-chain amino acids is an early event in
1202 human pancreatic adenocarcinoma development. *Nat Med* **20**, 1193-8 (2014).
- 1203 134. Mirus, J.E. et al. Cross-species antibody microarray interrogation identifies a 3-protein panel
1204 of plasma biomarkers for early diagnosis of pancreas cancer. *Clin Cancer Res* **21**, 1764-71
1205 (2015).
- 1206 135. Javeed, N. et al. Pancreatic Cancer-Derived Exosomes Cause Paraneoplastic beta-cell
1207 Dysfunction. *Clin Cancer Res* **21**, 1722-33 (2015).
- 1208 136. Cote, G.A. et al. A pilot study to develop a diagnostic test for pancreatic ductal
1209 adenocarcinoma based on differential expression of select miRNA in plasma and bile. *Am J*
1210 *Gastroenterol* **109**, 1942-52 (2014).
- 1211 137. Johnston, F.M. et al. Circulating mesothelin protein and cellular antimesothelin immunity in
1212 patients with pancreatic cancer. *Clin Cancer Res* **15**, 6511-8 (2009).

- 1213 138. Zhang, S. et al. Mesothelin virus-like particle immunization controls pancreatic cancer growth
1214 through CD8+ T cell induction and reduction in the frequency of CD4+ foxp3+ ICOS-
1215 regulatory T cells. *PLoS One* **8**, e68303 (2013).
- 1216 139. Wolfgang, C.L. et al. Recent progress in pancreatic cancer. *CA Cancer J Clin* **63**, 318-48 (2013).
- 1217 140. Harinck, F. et al. A multicentre comparative prospective blinded analysis of EUS and MRI for
1218 screening of pancreatic cancer in high-risk individuals. *Gut* (2015).
- 1219 141. Shin, E.J. et al. Linear-array EUS improves detection of pancreatic lesions in high-risk
1220 individuals: a randomized tandem study. *Gastrointest Endosc* (2015).
- 1221 142. Parikh, U., Marcus, C., Sarangi, R., Taghipour, M. & Subramaniam, R.M. FDG PET/CT in
1222 Pancreatic and Hepatobiliary Carcinomas: Value to Patient Management and Patient
1223 Outcomes. *PET Clin* **10**, 327-43 (2015).
- 1224 143. Poruk, K.E. et al. The clinical utility of CA 19-9 in pancreatic adenocarcinoma: diagnostic and
1225 prognostic updates. *Curr Mol Med* **13**, 340-51 (2013).
- 1226 144. Sah, R.P., Nagpal, S.J., Mukhopadhyay, D. & Chari, S.T. New insights into pancreatic cancer-
1227 induced paraneoplastic diabetes. *Nat Rev Gastroenterol Hepatol* **10**, 423-33 (2013).
- 1228 145. Hart, P.A. & Chari, S.T. Diabetes mellitus and pancreatic cancer: why the association matters?
1229 *Pancreas* **42**, 1207-9 (2013).
- 1230 146. Bettegowda, C. et al. Detection of Circulating Tumor DNA in Early- and Late-Stage Human
1231 Malignancies. *Sci Transl Med* **6**, 224ra24 (2014).
- 1232 147. Hruban, R.H., Pitman, M.B. & Klimstra, D.S. Tumors of the pancreas. Atlas of tumor pathology
1233 (American Registry of Pathology and Armed Forces Institute of Pathology, Washington, DC,
1234 2007).
- 1235 148. Jacobetz, M.A. et al. Hyaluronan impairs vascular function and drug delivery in a mouse
1236 model of pancreatic cancer. *Gut* (2012).
- 1237 149. Voong, K.R. et al. Resected pancreatic adenosquamous carcinoma: clinicopathologic review
1238 and evaluation of adjuvant chemotherapy and radiation in 38 patients. *Hum Pathol* **41**, 113-
1239 22 (2010).
- 1240 150. Le, D.T. et al. PD-1 Blockade in Tumors with Mismatch-Repair Deficiency. *N Engl J Med* **372**,
1241 2509-20 (2015).
- 1242 151. Basturk, O. et al. A Revised Classification System and Recommendations From the Baltimore
1243 Consensus Meeting for Neoplastic Precursor Lesions in the Pancreas. *Am J Surg Pathol* **39**,
1244 1730-41 (2015).
- 1245 152. Crippa, S. et al. Mucinous cystic neoplasm of the pancreas is not an aggressive entity: lessons
1246 from 163 resected patients. *Ann Surg* **247**, 571-9 (2008).
- 1247 153. Waters, J.A. et al. Survival after resection for invasive intraductal papillary mucinous
1248 neoplasm and for pancreatic adenocarcinoma: a multi-institutional comparison according to
1249 American Joint Committee on Cancer Stage. *J Am Coll Surg* **213**, 275-83 (2011).
- 1250 154. Mayo, S.C. et al. Management of patients with pancreatic adenocarcinoma: national trends
1251 in patient selection, operative management, and use of adjuvant therapy. *J Am Coll Surg* **214**,
1252 33-45 (2012).
- 1253 155. Chari, S.T. et al. Early detection of sporadic pancreatic cancer: summative review. *Pancreas*
1254 **44**, 693-712 (2015).
- 1255 156. Klein, A.P. et al. An absolute risk model to identify individuals at elevated risk for pancreatic
1256 cancer in the general population. *PLoS One* **8**, e72311 (2013).
- 1257 157. Risch, H.A., Yu, H., Lu, L. & Kidd, M.S. Detectable Symptomatology Preceding the Diagnosis of
1258 Pancreatic Cancer and Absolute Risk of Pancreatic Cancer Diagnosis. *Am J Epidemiol* **182**, 26-
1259 34 (2015).
- 1260 158. Klein, A.P. et al. Prospective risk of pancreatic cancer in familial pancreatic cancer kindreds.
1261 *Cancer Res* **64**, 2634-8 (2004).
- 1262 159. Al-Sukhni, W. et al. Screening for pancreatic cancer in a high-risk cohort: an eight-year
1263 experience. *J Gastrointest Surg* **16**, 771-83 (2012).

1264 160. Canto, M.I. et al. International Cancer of the Pancreas Screening (CAPS) Consortium summit
1265 on the management of patients with increased risk for familial pancreatic cancer. *Gut* **62**,
1266 339-47 (2013).

1267 161. Tanaka, M. et al. International consensus guidelines 2012 for the management of IPMN and
1268 MCN of the pancreas. *Pancreatol* **12**, 183-97 (2012).

1269 162. Pandharipande, P.V. et al. Targeted screening of individuals at high risk for pancreatic cancer:
1270 results of a simulation model. *Radiology* **275**, 177-87 (2015).

1271 163. Burris, H.A., 3rd et al. Improvements in survival and clinical benefit with gemcitabine as first-
1272 line therapy for patients with advanced pancreas cancer: a randomized trial. *J Clin Oncol* **15**,
1273 2403-13 (1997).

1274 164. Moore, M.J. et al. Erlotinib plus gemcitabine compared with gemcitabine alone in patients
1275 with advanced pancreatic cancer: a phase III trial of the National Cancer Institute of Canada
1276 Clinical Trials Group. *J Clin Oncol* **25**, 1960-6 (2007).

1277 165. Conroy, T. et al. FOLFIRINOX versus gemcitabine for metastatic pancreatic cancer. *N Engl J*
1278 *Med* **364**, 1817-25 (2011).

1279 166. Von Hoff, D.D. et al. Increased survival in pancreatic cancer with nab-paclitaxel plus
1280 gemcitabine. *N Engl J Med* **369**, 1691-703 (2013).

1281 167. Tempero, M. et al. Randomized phase II comparison of dose-intense gemcitabine: thirty-
1282 minute infusion and fixed dose rate infusion in patients with pancreatic adenocarcinoma. *J*
1283 *Clin Oncol* **21**, 3402-8 (2003).

1284 168. Golan, T. et al. Overall survival and clinical characteristics of pancreatic cancer in BRCA
1285 mutation carriers. *Br J Cancer* **111**, 1132-8 (2014).

1286 169. Cunningham, D. et al. Phase III randomized comparison of gemcitabine versus gemcitabine
1287 plus capecitabine in patients with advanced pancreatic cancer. *J Clin Oncol* **27**, 5513-8 (2009).

1288 170. Fine, R.L. et al. The gemcitabine, docetaxel, and capecitabine (GTX) regimen for metastatic
1289 pancreatic cancer: a retrospective analysis. *Cancer Chemother Pharmacol* **61**, 167-75 (2008).

1290 171. Chen, L.-T. et al. Expanded analyses of napoli-1: Phase 3 study of MM-398 (nal-IRI), with or
1291 without 5-fluorouracil and leucovorin, versus 5-fluorouracil and leucovorin, in metastatic
1292 pancreatic cancer (mPAC) previously treated with gemcitabine-based therapy. *ASCO Meeting*
1293 *Abstracts* **33**, 234 (2015).

1294 172. Hammel, P. et al. Comparison of chemoradiotherapy (CRT) and chemotherapy (CT) in
1295 patients with a locally advanced pancreatic cancer (LAPC) controlled after 4 months of
1296 gemcitabine with or without erlotinib: Final results of the international phase III LAP 07
1297 study. *Journal of Clinical Oncology* **31**, LBA4003 (2013).

1298 173. Hingorani, S.R. et al. High response rate and PFS with PEGPH20 added to nab-
1299 paclitaxel/gemcitabine in stage IV previously untreated pancreatic cancer patients with high-
1300 HA tumors: Interim results of a randomized phase II study. *ASCO Meeting Abstracts* **33**, 4006
1301 (2015).

1302 174. Hurwitz, H.I. et al. Randomized, Double-Blind, Phase II Study of Ruxolitinib or Placebo in
1303 Combination With Capecitabine in Patients With Metastatic Pancreatic Cancer for Whom
1304 Therapy With Gemcitabine Has Failed. *J Clin Oncol* (2015).

1305 175. Masso-Valles, D. et al. Ibrutinib exerts potent antifibrotic and antitumor activities in mouse
1306 models of pancreatic adenocarcinoma. *Cancer Res* **75**, 1675-81 (2015).

1307 176. Kleeff, J. et al. Pancreatic cancer microenvironment. *Int J Cancer* **121**, 699-705 (2007).

1308 177. Le, D.T. et al. Safety and survival with GVAX pancreas prime and Listeria Monocytogenes-
1309 expressing mesothelin (CRS-207) boost vaccines for metastatic pancreatic cancer. *J Clin Oncol*
1310 **33**, 1325-33 (2015).

1311 178. Soares, K.C. et al. PD-1/PD-L1 blockade together with vaccine therapy facilitates effector T-
1312 cell infiltration into pancreatic tumors. *J Immunother* **38**, 1-11 (2015).

1313 179. Khaled, Y.S., Wright, K., Melcher, A. & Jayne, D. Anti-cancer effects of oncolytic viral therapy
1314 combined with photodynamic therapy in human pancreatic cancer cell lines. *Lancet* **385**
1315 **Suppl 1**, S56 (2015).

- 1316 180. Xiao, Y. & Freeman, G.J. The microsatellite instable subset of colorectal cancer is a
1317 particularly good candidate for checkpoint blockade immunotherapy. *Cancer Discov* **5**, 16-8
1318 (2015).
- 1319 181. Bramhall, S.R. et al. Treatment and survival in 13,560 patients with pancreatic cancer, and
1320 incidence of the disease, in the West Midlands: an epidemiological study. *Br J Surg* **82**, 111-5
1321 (1995).
- 1322 182. van Heek, N.T. et al. Hospital volume and mortality after pancreatic resection: a systematic
1323 review and an evaluation of intervention in the Netherlands. *Ann Surg* **242**, 781-8, discussion
1324 788-90 (2005).
- 1325 183. Carpelan-Holmstrom, M. et al. Does anyone survive pancreatic ductal adenocarcinoma? A
1326 nationwide study re-evaluating the data of the Finnish Cancer Registry. *Gut* **54**, 385-7 (2005).
- 1327 184. Gudjonsson, B. Carcinoma of the pancreas: critical analysis of costs, results of resections, and
1328 the need for standardized reporting. *J Am Coll Surg* **181**, 483-503 (1995).
- 1329 185. Bilimoria, K.Y. et al. National failure to operate on early stage pancreatic cancer. *Ann Surg*
1330 **246**, 173-80 (2007).
- 1331 186. Amini, N., Spolverato, G., Kim, Y. & Pawlik, T.M. Trends in Hospital Volume and Failure to
1332 Rescue for Pancreatic Surgery. *J Gastrointest Surg* **19**, 1581-92 (2015).
- 1333 187. de Wilde, R.F. et al. Impact of nationwide centralization of pancreaticoduodenectomy on
1334 hospital mortality. *Br J Surg* **99**, 404-10 (2012).
- 1335 188. Hartwig, W. et al. Pancreatic cancer surgery in the new millennium: better prediction of
1336 outcome. *Ann Surg* **254**, 311-9 (2011).
- 1337 189. Neoptolemos, J.P. et al. Adjuvant chemotherapy with fluorouracil plus folinic acid vs
1338 gemcitabine following pancreatic cancer resection: a randomized controlled trial. *JAMA* **304**,
1339 1073-81 (2010).
- 1340 190. Neoptolemos, J.P. et al. A randomized trial of chemoradiotherapy and chemotherapy after
1341 resection of pancreatic cancer. *N Engl J Med* **350**, 1200-10 (2004).
- 1342 191. Oettle, H. et al. Adjuvant chemotherapy with gemcitabine and long-term outcomes among
1343 patients with resected pancreatic cancer: the CONKO-001 randomized trial. *JAMA* **310**, 1473-
1344 81 (2013).
- 1345 192. Cameron, J.L., Riall, T.S., Coleman, J. & Belcher, K.A. One thousand consecutive
1346 pancreaticoduodenectomies. *Ann Surg* **244**, 10-5 (2006).
- 1347 193. Diener, M.K. et al. Pylorus-preserving pancreaticoduodenectomy (pp Whipple) versus
1348 pancreaticoduodenectomy (classic Whipple) for surgical treatment of periampullary and
1349 pancreatic carcinoma. *Cochrane Database Syst Rev* **11**, CD006053 (2014).
- 1350 194. Hartwig, W. et al. Total pancreatectomy for primary pancreatic neoplasms: renaissance of an
1351 unpopular operation. *Ann Surg* **261**, 537-46 (2015).
- 1352 195. Mehrabi, A. et al. A systematic review and meta-analysis of laparoscopic versus open distal
1353 pancreatectomy for benign and malignant lesions of the pancreas: it's time to randomize.
1354 *Surgery* **157**, 45-55 (2015).
- 1355 196. Correa-Gallego, C. et al. Minimally-invasive vs open pancreaticoduodenectomy: systematic
1356 review and meta-analysis. *J Am Coll Surg* **218**, 129-39 (2014).
- 1357 197. Abdelgadir Adam, M. et al. Minimally Invasive Versus Open Pancreaticoduodenectomy for
1358 Cancer: Practice Patterns and Short-term Outcomes Among 7061 Patients. *Ann Surg* **262**,
1359 372-7 (2015).
- 1360 198. Cirocchi, R. et al. A systematic review on robotic pancreaticoduodenectomy. *Surg Oncol* **22**,
1361 238-46 (2013).
- 1362 199. Zureikat, A.H. et al. 250 robotic pancreatic resections: safety and feasibility. *Ann Surg* **258**,
1363 554-9; discussion 559-62 (2013).
- 1364 200. Tempero, M.A. et al. Pancreatic adenocarcinoma, version 2.2014: featured updates to the
1365 NCCN guidelines. *J Natl Compr Canc Netw* **12**, 1083-93 (2014).
- 1366 201. Bockhorn, M. et al. Borderline resectable pancreatic cancer: a consensus statement by the
1367 International Study Group of Pancreatic Surgery (ISGPS). *Surgery* **155**, 977-88 (2014).

- 1368 202. Yu, X.Z. et al. Benefit from synchronous portal-superior mesenteric vein resection during
1369 pancreaticoduodenectomy for cancer: a meta-analysis. *Eur J Surg Oncol* **40**, 371-8 (2014).
- 1370 203. Zhou, Y., Zhang, Z., Liu, Y., Li, B. & Xu, D. Pancreatectomy combined with superior mesenteric
1371 vein-portal vein resection for pancreatic cancer: a meta-analysis. *World J Surg* **36**, 884-91
1372 (2012).
- 1373 204. Mollberg, N. et al. Arterial resection during pancreatectomy for pancreatic cancer: a
1374 systematic review and meta-analysis. *Ann Surg* **254**, 882-93 (2011).
- 1375 205. Beane, J.D. et al. Distal pancreatectomy with celiac axis resection: what are the added risks?
1376 *HPB (Oxford)* **17**, 777-84 (2015).
- 1377 206. Smoot, R.L. & Donohue, J.H. Modified Appleby procedure for resection of tumors of the
1378 pancreatic body and tail with celiac axis involvement. *J Gastrointest Surg* **16**, 2167-9 (2012).
- 1379 207. Hartwig, W. et al. Multivisceral resection for pancreatic malignancies: risk-analysis and long-
1380 term outcome. *Ann Surg* **250**, 81-7 (2009).
- 1381 208. Tol, J.A. et al. Definition of a standard lymphadenectomy in surgery for pancreatic ductal
1382 adenocarcinoma: a consensus statement by the International Study Group on Pancreatic
1383 Surgery (ISGPS). *Surgery* **156**, 591-600 (2014).
- 1384 209. Dasari, B.V. et al. Extended Versus Standard Lymphadenectomy for Pancreatic Head Cancer:
1385 Meta-Analysis of Randomized Controlled Trials. *J Gastrointest Surg* **19**, 1725-32 (2015).
- 1386 210. Michalski, C.W. et al. Systematic review and meta-analysis of standard and extended
1387 lymphadenectomy in pancreaticoduodenectomy for pancreatic cancer. *Br J Surg* **94**, 265-73
1388 (2007).
- 1389 211. Gleisner, A.L. et al. Is resection of periampullary or pancreatic adenocarcinoma with
1390 synchronous hepatic metastasis justified? *Cancer* **110**, 2484-92 (2007).
- 1391 212. Michalski, C.W. et al. Resection of primary pancreatic cancer and liver metastasis: a
1392 systematic review. *Dig Surg* **25**, 473-80 (2008).
- 1393 213. Arnaoutakis, G.J. et al. Pulmonary resection for isolated pancreatic adenocarcinoma
1394 metastasis: an analysis of outcomes and survival. *J Gastrointest Surg* **15**, 1611-7 (2011).
- 1395 214. Gillen, S., Schuster, T., Friess, H. & Kleeff, J. Palliative resections versus palliative bypass
1396 procedures in pancreatic cancer--a systematic review. *Am J Surg* **203**, 496-502 (2012).
- 1397 215. Chandrasegaram, M.D. et al. Meta-analysis of radical resection rates and margin assessment
1398 in pancreatic cancer. *Br J Surg* (2015).
- 1399 216. Esposito, I. et al. Most pancreatic cancer resections are R1 resections. *Ann Surg Oncol* **15**,
1400 1651-60 (2008).
- 1401 217. Schlitter, A.M. & Esposito, I. Definition of microscopic tumor clearance (r0) in pancreatic
1402 cancer resections. *Cancers (Basel)* **2**, 2001-10 (2010).
- 1403 218. Traverso, L.W. Pancreatic cancer: surgery alone is not sufficient. *Surg Endosc* **20 Suppl 2**,
1404 S446-9 (2006).
- 1405 219. Regine, W.F. et al. Fluorouracil-based chemoradiation with either gemcitabine or fluorouracil
1406 chemotherapy after resection of pancreatic adenocarcinoma: 5-year analysis of the U.S.
1407 Intergroup/RTOG 9704 phase III trial. *Ann Surg Oncol* **18**, 1319-26 (2011).
- 1408 220. Gillen, S., Schuster, T., Meyer Zum Buschenfelde, C., Friess, H. & Kleeff, J.
1409 Preoperative/neoadjuvant therapy in pancreatic cancer: a systematic review and meta-
1410 analysis of response and resection percentages. *PLoS Med* **7**, e1000267 (2010).
- 1411 221. Heinrich, S. et al. Adjuvant gemcitabine versus NEOadjuvant gemcitabine/oxaliplatin plus
1412 adjuvant gemcitabine in resectable pancreatic cancer: a randomized multicenter phase III
1413 study (NEOPAC study). *BMC Cancer* **11**, 346 (2011).
- 1414 222. Tachezy, M. et al. Sequential neoadjuvant chemoradiotherapy (CRT) followed by curative
1415 surgery vs. primary surgery alone for resectable, non-metastasized pancreatic
1416 adenocarcinoma: NEOPA- a randomized multicenter phase III study (NCT01900327,
1417 DRKS00003893, ISRCTN82191749). *BMC Cancer* **14**, 411 (2014).

- 1418 223. Ferrone, C.R. et al. Radiological and surgical implications of neoadjuvant treatment with
1419 FOLFIRINOX for locally advanced and borderline resectable pancreatic cancer. *Ann Surg* **261**,
1420 12-7 (2015).
- 1421 224. Nitsche, U. et al. Resectability After First-Line FOLFIRINOX in Initially Unresectable Locally
1422 Advanced Pancreatic Cancer: A Single-Center Experience. *Ann Surg Oncol* (2015).
- 1423 225. Cancer-Research-UK-Liverpool-Cancer-Trials-Unit. ESPAC-5F: European Study group for
1424 Pancreatic Cancer - Trial 5F. www.isrctn.com/ISRCTN89500674 (2014).
- 1425 226. Katz, M.H.G. et al. Preoperative modified FOLFIRINOX (mFOLFIRINOX) followed by
1426 chemoradiation (CRT) for borderline resectable (BLR) pancreatic cancer (PDAC): Initial results
1427 from Alliance Trial A021101. *Journal of Clinical Oncology* **33** (2015).
- 1428 227. Pawlik, T.M. et al. Evaluating the impact of a single-day multidisciplinary clinic on the
1429 management of pancreatic cancer. *Ann Surg Oncol* **15**, 2081-8 (2008).
- 1430 228. Rainbird, K., Perkins, J., Sanson-Fisher, R., Rolfe, I. & Anseline, P. The needs of patients with
1431 advanced, incurable cancer. *Br J Cancer* **101**, 759-64 (2009).
- 1432 229. Aaronson, N.K. et al. The European Organization for Research and Treatment of Cancer QLQ-
1433 C30: a quality-of-life instrument for use in international clinical trials in oncology. *J Natl*
1434 *Cancer Inst* **85**, 365-76 (1993).
- 1435 230. Cella, D.F. et al. The Functional Assessment of Cancer Therapy scale: development and
1436 validation of the general measure. *J Clin Oncol* **11**, 570-9 (1993).
- 1437 231. Fitzsimmons, D. et al. Development of a disease specific quality of life (QoL) questionnaire
1438 module to supplement the EORTC core cancer QoL questionnaire, the QLQ-C30 in patients
1439 with pancreatic cancer. EORTC Study Group on Quality of Life. *Eur J Cancer* **35**, 939-41 (1999).
- 1440 232. Cella, D. et al. Validity of the FACT Hepatobiliary (FACT-Hep) questionnaire for assessing
1441 disease-related symptoms and health-related quality of life in patients with metastatic
1442 pancreatic cancer. *Qual Life Res* **22**, 1105-12 (2013).
- 1443 233. Serrano, P.E. et al. Quality of life in a prospective, multicenter phase 2 trial of neoadjuvant
1444 full-dose gemcitabine, oxaliplatin, and radiation in patients with resectable or borderline
1445 resectable pancreatic adenocarcinoma. *Int J Radiat Oncol Biol Phys* **90**, 270-7 (2014).
- 1446 234. Short, M. et al. Impact of gemcitabine chemotherapy and 3-dimensional conformal radiation
1447 therapy/5-fluorouracil on quality of life of patients managed for pancreatic cancer. *Int J*
1448 *Radiat Oncol Biol Phys* **85**, 157-62 (2013).
- 1449 235. Huang, J.J. et al. Quality of life and outcomes after pancreaticoduodenectomy. *Ann Surg* **231**,
1450 890-8 (2000).
- 1451 236. Reni, M. et al. Quality of life assessment in advanced pancreatic adenocarcinoma: results
1452 from a phase III randomized trial. *Pancreatolgy* **6**, 454-63 (2006).
- 1453 237. Hurt, C.N. et al. Health-Related Quality of Life in SCALOP, a Randomized Phase 2 Trial
1454 Comparing Chemoradiation Therapy Regimens in Locally Advanced Pancreatic Cancer.
1455 *International Journal of Radiation Oncology • Biology • Physics* **93**, 810-818 (2015).
- 1456 238. Gaertner, J. et al. The Role of End-of-Life Issues in the Design and Reporting of Cancer Clinical
1457 Trials: A Structured Literature Review. *PLoS One* **10**, e0136640 (2015).
- 1458 239. Biankin, A.V., Piantadosi, S. & Hollingsworth, S.J. Patient-centric trials for therapeutic
1459 development in precision oncology. *Nature* **526**, 361-70 (2015).
- 1460 240. Chang, D.K., Grimmond, S.M., Evans, T.R.J. & Biankin, A.V. Mining the Genomes of
1461 Exceptional Responders. *Nature Rev Cancer* **14** (2014).
- 1462 241. \$250, P.G.T.W.S.a.N.L.P.o. [http://www.scientificamerican.com/article/craig-venter-s-](http://www.scientificamerican.com/article/craig-venter-s-company-in-deal-for-whole-exome-tests-at-new-low-cost/)
1463 [company-in-deal-for-whole-exome-tests-at-new-low-cost/](http://www.scientificamerican.com/article/craig-venter-s-company-in-deal-for-whole-exome-tests-at-new-low-cost/) (2015).
- 1464 242. 2015, P.C.W.P. <http://www.euapm.eu/news,86.html> (2015).
- 1465 243. Dindo, D., Demartines, N. & Clavien, P.A. Classification of surgical complications: a new
1466 proposal with evaluation in a cohort of 6336 patients and results of a survey. *Ann Surg* **240**,
1467 205-13 (2004).
- 1468 244. Bassi, C. et al. Postoperative pancreatic fistula: an international study group (ISGPF)
1469 definition. *Surgery* **138**, 8-13 (2005).

- 1470 245. Wente, M.N. et al. Delayed gastric emptying (DGE) after pancreatic surgery: a suggested
1471 definition by the International Study Group of Pancreatic Surgery (ISGPS). *Surgery* **142**, 761-8
1472 (2007).
- 1473 246. Wente, M.N. et al. Postpancreatectomy hemorrhage (PPH): an International Study Group of
1474 Pancreatic Surgery (ISGPS) definition. *Surgery* **142**, 20-5 (2007).
- 1475 247. Common Terminology Criteria for Adverse Events (CTCAE), V.
1476 [http://ctep.cancer.gov/protocolDevelopment/electronic_applications/docs/ctcv20_4-30-](http://ctep.cancer.gov/protocolDevelopment/electronic_applications/docs/ctcv20_4-30-992.pdf)
1477 [992.pdf](http://ctep.cancer.gov/protocolDevelopment/electronic_applications/docs/ctcv20_4-30-992.pdf). (2010).
- 1478 248. Giardiello, F.M. et al. Very high risk of cancer in familial Peutz-Jeghers syndrome.
1479 *Gastroenterology* **119**, 1447-53 (2000).
- 1480 249. Lowenfels, A.B. et al. Hereditary pancreatitis and the risk of pancreatic cancer. International
1481 Hereditary Pancreatitis Study Group. *J Natl Cancer Inst* **89**, 442-6 (1997).
- 1482 250. Howes, N. et al. Clinical and genetic characteristics of hereditary pancreatitis in Europe. *Clin*
1483 *Gastroenterol Hepatol* **2**, 252-61 (2004).
- 1484 251. Goldstein, A.M. et al. Increased risk of pancreatic cancer in melanoma-prone kindreds with
1485 p16INK4 mutations. *N Engl J Med* **333**, 970-4 (1995).
- 1486 252. Borg, A. et al. High frequency of multiple melanomas and breast and pancreas carcinomas in
1487 CDKN2A mutation-positive melanoma families. *J Natl Cancer Inst* **92**, 1260-6 (2000).
- 1488 253. Kastrinos, F. et al. Risk of pancreatic cancer in families with Lynch syndrome. *JAMA* **302**,
1489 1790-5 (2009).
- 1490 254. Maisonneuve, P., Marshall, B.C. & Lowenfels, A.B. Risk of pancreatic cancer in patients with
1491 cystic fibrosis. *Gut* **56**, 1327-8 (2007).
- 1492 255. Giardiello, F.M. et al. Increased risk of thyroid and pancreatic carcinoma in familial
1493 adenomatous polyposis. *Gut* **34**, 1394-6 (1993).
- 1494 256. Iqbal, J. et al. The incidence of pancreatic cancer in BRCA1 and BRCA2 mutation carriers. *Br J*
1495 *Cancer* **107**, 2005-9 (2012).
- 1496 257. Breast Cancer Linkage, C. Cancer risks in BRCA2 mutation carriers. *J Natl Cancer Inst* **91**,
1497 1310-6 (1999).
- 1498 258. Ruijs, M.W. et al. TP53 germline mutation testing in 180 families suspected of Li-Fraumeni
1499 syndrome: mutation detection rate and relative frequency of cancers in different familial
1500 phenotypes. *J Med Genet* **47**, 421-8 (2010).

1501